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Chapter 1

Introduction

Most linear programming lectures take the following introduction and definition as the start of the course. Let $M \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{c} \in \mathbb{R}^n$; consider the standard primal linear program in an inequality form:

$$\begin{aligned} \max_{\mathbf{x} \in \mathbb{R}^n} \quad & \langle \mathbf{c}, \mathbf{x} \rangle \\ \text{subject to} \quad & M\mathbf{x} \leq \mathbf{b}. \end{aligned}$$

This compact form expands component-wise to the familiar system of m linear inequalities

$$m_{i,1}x_1 + m_{i,2}x_2 + \cdots + m_{i,n}x_n \leq b_i,$$

where $m_{i,j}$ is the (i, j) -th entry of M , and b_j and x_j are the entry on the i -th coordinate of $\mathbf{b}$ and $\mathbf{x}$, respectively, for $i \in \{1, 2, \dots, m\}$ and $j \in \{1, 2, \dots, n\}$. Many textbooks further assume $\mathbf{x} \geq \mathbf{0}$ and, by a seemingly symmetrical rule, introduce a vector $\mathbf{y} \in \mathbb{R}^m$ of dual variables to define the dual problem as

$$\begin{aligned} \min \quad & \langle \mathbf{y}, \mathbf{b} \rangle \\ \text{subject to} \quad & M^\top \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y} \geq \mathbf{0}. \end{aligned}$$

The connection between these two problems is often justified *a posteriori* by the following: for any feasible $\mathbf{x}$ and $\mathbf{y}$, i.e., $M\mathbf{x} \leq \mathbf{b}$ and $M^\top \mathbf{y} \geq \mathbf{c}$, we have

$$\langle \mathbf{c}, \mathbf{x} \rangle \leq \langle M^\top \mathbf{y}, \mathbf{x} \rangle = \langle \mathbf{y}, M\mathbf{x} \rangle \leq \langle \mathbf{y}, \mathbf{b} \rangle.$$

This chain proves weak duality, and therefore every feasible dual solution gives an upper bound for the primal.

However, there is a logic gap: the system of inequalities $M^\top \mathbf{y} \geq \mathbf{c}$ in the dual problem is introduced because the chain of inequalities holds. While this verifies the duality relationship, it does not explain where the dual variables come from, why the signs are

what they are, or how to generalize the construction of other constraints.

In this booklet, we take a pedagogical route. With only a primal problem in linear optimization, we first aim to derive its dual from Farkas' Lemma. On this path, we discover geometric views of optimization from different angles, which leads to the Minkowski-Weyl Theorem and, then, the characterizations for “vertices” for linear optimization problems and feasible solutions from implied linear algebra from the constraint system. We set our first milestone as George Dantzig's simplex method, a staple of undergraduate courses that is frequently taught as a mere mechanical pivoting procedure, masking the rich geometric and theoretical machinery that actually powers it.

To maintain a focused, linear reading experience, we adhere to two stylistic conventions. First, we try to be minimalist. With a primary, simple, and sole target of understanding Dantzig's simplex method for linear programs, we detour as little as possible before achieving this target and define terms only when they are necessary. Second, we use mathematical terms to reflect pedagogical weight rather than abstract depth. We use the term *proposition*, with or without a proof, as facts whose full proofs lie outside our scope. *Theorems* form the core narrative arc; when presented without a proof, the verification is straightforward and left as a constructive exercise.

In many texts, the dot product between two elements in $\mathbb{R}^n$, n -dimensional real vectors, is denoted by $\mathbf{x}^\top \mathbf{y}$ or $\langle \mathbf{x}, \mathbf{y} \rangle$; we adopt the bracket notation $\langle \mathbf{x}, \mathbf{y} \rangle$. First, it spares us from explicitly invoking the transpose operation and the associated matrix multiplication, which saves mental power when discussing associativity, such as $\langle \mathbf{x}, A\mathbf{y} \rangle = \langle A^\top \mathbf{x}, \mathbf{y} \rangle$. Moreover, when readers arrive at $\langle \cdot, \cdot \rangle$, we automatically parse it as a real number instead of a matrix multiplication with some hidden dimension to discover. Finally, the last author believes that the notation $\mathbf{x}^\top \mathbf{y}$ assumes the operation on matrices of the vectors $\mathbf{x}$ and $\mathbf{y}$, and after transposing and multiplication, it gives a 1-by-1 matrix instead of a scalar, in which they are interchangeable but would be more reader-friendly if the scalar-property is addressed explicitly.

For this note, we assume a working familiarity with introductory linear algebra and basic real analysis in Euclidean space. In linear algebra, we expect readers to have internalized the concepts of linear transformations, linear independence, orthogonality, and bases of a subspace. In real analysis, we assume readers have internalized the equivalence between different definitions of open and closed sets in Euclidean spaces, understood the definition of the interior, boundary, and closure of a set, and have understood the equivalence between Cauchy-ness and convergence in a closed set. Because our focus is geometric intuition within Euclidean space, we keep real-analytical machineries to a minimum, occasionally omitting technical measure-theoretic or topological details when standard geometric intuition suffices.

This chapter collects tools in real analysis, convex geometry, and cone theory that form the bedrock of this booklet. Rather than re-deriving these results from first principles, we state these core results as propositions. One may think of Chapter 1 as a self-diagnostic baseline: memorizing their proofs is not necessary, but one should feel comfortable reading,

visualizing, and applying them with guides and thus be well-prepared.

1.1 Real Analysis Preliminaries

apter: prelim)? We primarily deal with the vector spaces $\mathbb{R}^n$ with $n \geq 1$ and their subsets. We denote the boundary and the closure of a set S by $\text{bd}(S)$ and $\text{cl}(S)$, respectively. We take the following analytical fact as given.

sed intersection) **Proposition 1.1.** *Every finite or infinite intersection of closed sets is closed.* $\square$

We denote by $\mathbb{R}$ the set of real numbers. We denote the set of non-negative real numbers as $\mathbb{R}_+$ and the set of positive real numbers as $\mathbb{R}_{++}$; similarly, we denote the set of all non-positive (resp. negative) real numbers as $\mathbb{R}_-$ (resp. $\mathbb{R}_{--}$). We denote and define the set of all (entry-wise) non-negative vectors as

$$\mathbb{R}_+^n = \left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} : \forall i \in \{1, 2, \dots, n\}, x_i \geq 0 \right\}$$

and the set of all (entry-wise) non-positive vectors

$$\mathbb{R}_-^n = \left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} : \forall i \in \{1, 2, \dots, n\}, x_i \leq 0 \right\}.$$

In the context of linear optimization, all the comparisons between two vectors are entry-wise; that is, for two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, where x_i and y_i denotes the i -th entry of $\mathbf{x}$ and $\mathbf{y}$, respectively, we denote $\mathbf{x} \leq \mathbf{y}$ if $x_i \leq y_i$ for all $i \in \{1, 2, \dots, n\}$. For elements $\mathbf{x} \in \mathbb{R}_+^n$ and $\mathbf{y} \in \mathbb{R}_-^n$, we may write that $\mathbf{y} \leq \mathbf{0} \leq \mathbf{x}$.

Proposition 1.2. *Let $\mathbf{y}, \mathbf{b}_1, \mathbf{b}_2 \in \mathbb{R}^m$. If $\mathbf{b}_1 \leq \mathbf{b}_2$ and $\mathbf{y} \geq \mathbf{0}$, then*

$$\langle \mathbf{y}, \mathbf{b}_1 \rangle \leq \langle \mathbf{y}, \mathbf{b}_2 \rangle.$$

Proof: Rewrite $\mathbf{b}_1 \leq \mathbf{b}_2$ as $\mathbf{b}_2 - \mathbf{b}_1 \geq \mathbf{0}$. As $\mathbf{y} \geq \mathbf{0}$ and every term we are dealing with is nonnegative, we have that $\langle \mathbf{y}, \mathbf{b}_2 - \mathbf{b}_1 \rangle \geq 0$, which gives that $\langle \mathbf{y}, \mathbf{b}_2 - \mathbf{b}_1 \rangle = \langle \mathbf{y}, \mathbf{b}_2 \rangle - \langle \mathbf{y}, \mathbf{b}_1 \rangle \geq 0$ and thus $\langle \mathbf{y}, \mathbf{b}_2 \rangle \geq \langle \mathbf{y}, \mathbf{b}_1 \rangle$. $\square$

For inline context, given $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{x}' \in \mathbb{R}^m$, we may write $(\mathbf{x}, \mathbf{x}') \in \mathbb{R}^{m+n}$ as the concatenation of the two vectors; this notation coincides with the notation for an open

real number interval, so we will explicitly explain before using this notation.

1.2 Convex Sets

A set C , where $C \subseteq \mathbb{R}^n$, is *convex* if for all $\mathbf{x}, \mathbf{y} \in C$ and $\alpha \in [0, 1]$, we have that $\alpha\mathbf{x} + (1 - \alpha)\mathbf{y} \in C$. The closure and the interior of a convex set are also convex. A vector $\mathbf{z}$ has a linear combination of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ if there are $a_1, a_2, \dots, a_m \in \mathbb{R}$ so that $\mathbf{z} = a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_m\mathbf{x}_m$. The linear combination is a *conic combination* if the coefficients are non-negative; that is, $a_i \geq 0$ for every $i \in \{1, 2, \dots, m\}$. The linear combination is a *convex combination* if it is a conic combination and $\sum_{i=1}^m a_i = 1$.

Definition 1.3. A function $f : C \rightarrow \mathbb{R}$, where $C \subseteq \mathbb{R}^n$, is *convex* if for all $\mathbf{x}, \mathbf{y} \in C$ and $\alpha \in [0, 1]$, we have that

$$f(\alpha\mathbf{x} + (1 - \alpha)\mathbf{y}) \leq \alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y}).$$

Let C and S be sets, where $C \subseteq S \subseteq \mathbb{R}^n$, and f be a function that $f : S \rightarrow \mathbb{R}$. Suppose that f is not convex; then we say f is *convex over C* if $f|_C$ is convex.

We take the following analytical facts for convex sets as given.

convex intersection

Proposition 1.4. *Every finite or infinite intersection of convex sets is convex.* $\square$

Proposition 1.5. *Let C be a convex subset of $\mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function. We have that*

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \mathbf{y} - \mathbf{x}, \nabla f(\mathbf{x}) \rangle$$

for all $\mathbf{x}, \mathbf{y} \in C$. $\square$

Proposition 1.6. *Let C be a convex subset of $\mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a twice-differentiable function. If $\nabla^2 f(\mathbf{x})$ is positive semidefinite for all $\mathbf{x} \in C$, then f is convex over C . If C is open and f is convex over C , then $\nabla^2 f(\mathbf{x})$ is positive semidefinite for all $\mathbf{x} \in C$. $\square$*

The *convex hull* of a set X , denoted by $\text{conv}(X)$, is the minimal convex set that contains X ; alternatively, $\text{conv}(X)$ is the intersection of all convex sets containing X .

plane separation)

Proposition 1.7 (Strict Hyperplane Separation Theorem). *Suppose $C, D \subseteq \mathbb{R}^n$ are two closed convex sets, with D compact and $C \cap D = \emptyset$. There then is a point $\mathbf{a} \in \mathbb{R}^n, q \in \mathbb{R}$ such that $\mathbf{a} \neq \mathbf{0}$ and*

$$\forall \mathbf{x} \in C, \langle \mathbf{a}, \mathbf{x} \rangle \geq q \quad \text{and} \quad \forall \mathbf{x} \in D, \langle \mathbf{a}, \mathbf{x} \rangle < q. \quad \square$$

Since a singleton is closed and compact, we have the following corollary.

Corollary 1.8. *If $\mathbf{b} \in \mathbb{R}^n$ is a point such that $\mathbf{b} \notin C$, where $C \subseteq \mathbb{R}^n$, then there is a hyperplane, defined by $\mathbf{a} \in \mathbb{R}^n$ and $q \in \mathbb{R}$ that separates $\langle \mathbf{a}, \mathbf{b} \rangle > q$ and $\langle \mathbf{a}, \mathbf{c} \rangle < q$ for all $\mathbf{c} \in C$.* $\square$

1.3 Geometric Cones

A set C , where $C \subseteq \mathbb{R}^n$ is a *cone* if for all $\mathbf{x} \in C$ and $c \in \mathbb{R}_+$, we have that $c\mathbf{x} \in C$. A cone need not be convex, as the boundary of a cone is also a cone. For example, $\mathbb{R}_+^n$ is a cone; its boundary, consisting of the non-negative vectors with at least one zero-entry, is a cone but not convex.

The *cone generated by a set X* , where $X = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\} \subseteq \mathbb{R}^n$, is denoted and defined as the set of conic combinations of X ,

$$\text{cone}(X) = \{a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_m\mathbf{x}_m : \forall i \in \{1, 2, \dots, m\}, a_i \in \mathbb{R}_+, \mathbf{x}_i \in X\};$$

Clearly, $\mathbf{0} \in \text{cone}(X)$ for any X . The *polar cone* of a cone C is denoted and defined as $C^\circ = \{y \in \mathbb{R}^n : \forall x \in C, \langle y, x \rangle \leq 0\}$. We take the following facts on cones as given.

(thm: bipolar)

Proposition 1.9 (The Bipolar Theorem). *For a nonempty set X , we have*

$$X^\circ = (\text{cl}(X))^\circ = (\text{conv}(X))^\circ = (\text{cone}(X))^\circ.$$

Further, given a cone C , we have that

$$(C^\circ)^\circ = \text{cl}(\text{conv}(C))$$

and, thus, $C = (C^\circ)^\circ$ if the cone C is already closed and convex. $\square$

The next theorem is every math student's favorite type of theorem: it states that a vector in a cone in $\mathbb{R}^n$ can be represented by n or fewer vectors, and a vector in a convex hull can be represented by $n + 1$ or fewer vectors.

-aff combination

Proposition 1.10. *Let X be a non-empty subset of $\mathbb{R}^n$; we may assume that X is a linearly independent set.*

1. *Every nonzero $x \in \text{cone}(X)$ can be represented as a conic combination of linearly independent vectors in X .*
2. *Every $x \in \text{conv}(X) \setminus X$ can be represented as a convex combination of m vectors $x_1, x_2, \dots, x_m \in X$ where $x_2 - x_1, x_3 - x_1, \dots, x_m - x_1$ are linearly independent.*

□

The first statement is clear as a basic linear-algebraic argument about dimensionality. For the second part, consider $X = \{\mathbf{x}_1 = (1, 0)^\top, \mathbf{x}_2 = (0, 1)^\top, \mathbf{x}_3 = (-1, -1)^\top\} \subseteq \mathbb{R}^2$. Clearly, $\mathbf{0} \in \text{conv}(X)$ but $\mathbf{0} = \frac{1}{3}\mathbf{x}_1 + \frac{1}{3}\mathbf{x}_2 + \frac{1}{3}\mathbf{x}_3$ is a convex combination and X is not linear independent. As in the second point of Proposition 1.10, a list of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ is said to be *affinely independent* if the list of $m - 1$ vectors $\mathbf{x}_2 - \mathbf{x}_1, \mathbf{x}_3 - \mathbf{x}_1, \dots, \mathbf{x}_m - \mathbf{x}_1$ is linearly independent. Recall that, suppose $V \subseteq \mathbb{R}^n$ is a list of m linearly independent vectors, then $\dim(\text{span}(V)) = m$; we have a similar dimensionality argument using affine independence.

Consider a line segment L with endpoints $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ for $n \geq 2$. There is a unique line that contains L , and obviously, we can talk about $L \setminus \{\mathbf{a}, \mathbf{b}\}$ as the “internal part” of L . However, the interior of L is empty, as every point in L has its delta-ball intersecting something not on the line segment. The concept of affine set and relative interior precisely tackles this issue.

An *affine set* is a translated subspace, that is, let S be a subspace of $\mathbb{R}^n$ and $\mathbf{x} \in \mathbb{R}^n$, a set A is an affine set if it has a format of $A = \mathbf{x} + S = \{\mathbf{x} + \mathbf{s} : \mathbf{s} \in S\}$; the subspace S is also known as the *subspace parallel to A* . Given an affine set A , the subspace parallel to it is unique. Given a set $X \subseteq \mathbb{R}^n$, the *affine hull of X* , denoted by $\text{aff}(X)$, is the minimal affine set that contains X . Alternatively, $\text{aff}(X)$ is the intersection of all affine sets in $\mathbb{R}^n$ that contain X . The dimension of $\text{aff}(X)$ is the greatest number of affinely independent vectors in X .

The *relative interior* of a set X is the interior of X in $\text{aff}(X)$, denoted by $\text{ri}(X)$. For the aforementioned line segment L example, the line that properly contains L is precisely the affine hull $\text{aff}(L)$ and $\text{ri}(L) = L \setminus \{\mathbf{a}, \mathbf{b}\}$. We take the following face as given.

prop: nonempty ri

Proposition 1.11. *The relative interior of a convex set is nonempty.*

Proof: Let K be a convex set. For the case when $K = \{\mathbf{x}\}$ is a singleton, then $\text{aff}(K) = K$ is a single point. By definition, K is an open set such that every point has a distance of 0 from $\mathbf{x}$; thus, $x \in \text{ri}(K)$.

Suppose K contains one or more affine-independent vectors, that is, there are at least two distinct vectors $\mathbf{x}_1, \mathbf{x}_2 \in K$. We then have $\frac{1}{2}\mathbf{x}_1 + \frac{1}{2}\mathbf{x}_2 \in \text{ri}(K)$ by the definition of a convex set and relative interior. □

Chapter 2

Linear Optimization and its Dual

`lin opt dual`? We start this chapter by introducing a standard textbook-style linear program, (P0). We then investigate the celebrated Farkas' Lemma, Theorem 2.1, to give a first characterization for the feasibility of a linear program. Farkas' Lemma is sufficient for proving the strong duality of a linear program; therefore, we prove a specific case for the minimax theorem in Theorem 2.8; we then generalize it to a cone-version of the statement, in which the proof is straightforward. This then provides enough motivation to properly define the dual of a primal problem.

In the second part of the chapter, we move to a geometric point of view for Farkas' Lemma, Theorem 2.13. The two versions of Farkas' Lemma are also known as the H-representation and V-representation of a polyhedral set, which we will then further investigate in Chapter 3.

2.1 Farkas' Lemma and Dual Problem

In linear optimization problems, the constraints are expressed as linear transformations; we may exploit the bounds in inner product spaces to obtain an upper bound on the maximum. We first restate the optimization problem. Consider the system of linear inequalities and the optimization problem that

$$\begin{array}{ll} \min_{\mathbf{x} \in \mathbb{R}^n} & \langle \mathbf{c}, \mathbf{x} \rangle \\ \text{subject to} & M\mathbf{x} \leq \mathbf{b}. \end{array} \quad (\text{P0}) \quad \boxed{\text{P0}}$$

and we may view M as a linear transformation $M : \mathbb{R}^n \rightarrow \mathbb{R}^m$; in a way, the optimization problem (P0) is to study the row space of the matrix M , where one needs to find a vector in $\mathbb{R}^n$ so that it has a direction close to the objective $\mathbf{c}$ and obeys the constraint M .

Before looking for solutions to the linear program (P0), we first want to check whether solutions exist. The following theorem shows that the feasibility of a linear system is completely determined by an associated dual system in $\mathbb{R}^m$. Recall that the hyperplane separation theorem (Theorem 1.7) states the following: if a point $\mathbf{b}$ does not lie inside a

closed convex set Z , you can always draw a hyperplane that completely separates $\mathbf{b}$ from Z and does not touch $\mathbf{b}$ at all.

Theorem 2.1 (Farkas' Lemma). *Let $M \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$ and $x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m$ be a list of variables written in their vector forms $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{u} \in \mathbb{R}^m$. Consider the following two systems of linear inequalities.*

$$M\mathbf{x} \leq \mathbf{b} \tag{2.2} \text{eqn: P}$$

and

$$\begin{aligned} M^\top \mathbf{u} &= 0 \\ \mathbf{u} &\geq 0 \\ \langle \mathbf{u}, \mathbf{b} \rangle &< 0. \end{aligned} \tag{2.3} \text{eqn: D}$$

The system of inequalities (2.2) has a feasible solution if and only if the system of equalities and inequalities (2.3) has no solution.

Proof: The forward direction, suppose that there are $\mathbf{x}$ and $\mathbf{u}$ so that (2.2) and (2.3) both have solutions; we have that

$$0 > \langle \mathbf{u}, \mathbf{b} \rangle \geq \langle \mathbf{u}, M\mathbf{x} \rangle = \langle M^\top \mathbf{u}, \mathbf{x} \rangle = 0,$$

which leads to a contradiction and, thus, (2.3) has no solution. For the converse direction, we proceed via contrapositive, supposing (2.2) has no solution; we will show that (2.3) has a solution.

The assumption that (2.2) has no solution means the set $\{M\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}$ does not include $\mathbf{b}$; since, otherwise, we would have a solution $\mathbf{x}^*$ so that $M\mathbf{x}^* = \mathbf{b} \leq \mathbf{b}$. In particular, we have that $Z = \{M\mathbf{x} + \mathbf{s} : \mathbf{x} \in \mathbb{R}^n, \mathbf{s} \in \mathbb{R}_+^m\}$ does not include $\mathbf{b}$. By the *hyperplane separation theorem*, there exists $\mathbf{u}^* \in \mathbb{R}^m$ and $q \in \mathbb{R}$ such that $\langle \mathbf{u}^*, \mathbf{b} \rangle < q$ and $\langle \mathbf{u}^*, \mathbf{z} \rangle \geq q$ for every $\mathbf{z} \in Z$. In particular, since M is a linear transformation, $\mathbf{x}$ is taken over $\mathbb{R}^n$, $\mathbf{b} \notin \text{Im}(M) := \{\mathbf{y} \in \mathbb{R}^m : \mathbf{y} = M\mathbf{x}\}$, and $\text{Im}(M)$ is a subspace of $\mathbb{R}^m$, we have that Z is a convex cone and we may take $q = 0$.

We now verify that $\mathbf{u}^*$ is a solution to (2.3). First, we have $\langle \mathbf{u}^*, \mathbf{b} \rangle < 0$ by the choice according to the separation theorem. By the construction of Z , we may say for all $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{s} \in \mathbb{R}_+^m$,

$$\langle \mathbf{u}^*, M\mathbf{x} + \mathbf{s} \rangle = \langle \mathbf{u}^*, M\mathbf{x} \rangle + \langle \mathbf{u}^*, \mathbf{s} \rangle \geq 0, \tag{2.4} \text{eqn: construct.}$$

where the inequality is from the separation theorem. When we fix $\mathbf{x} = 0$, Inequality (2.4) gives that $\langle \mathbf{u}^*, \mathbf{s} \rangle \geq 0$ for any $\mathbf{s} \in \mathbb{R}_+^m$: we may conclude that $\mathbf{u}^* \in \mathbb{R}_+^m$ (i.e., $\mathbf{y}^* \geq 0$).

Finally, we assume $\mathbf{s} = 0$, we have that $\langle \mathbf{u}^*, M\mathbf{x} \rangle + \langle \mathbf{u}^*, \mathbf{s} \rangle = \langle \mathbf{u}^*, M\mathbf{x} \rangle = \langle M^\top \mathbf{u}^*, \mathbf{x} \rangle \geq 0$. Now, by properties of inner products, for every $\mathbf{x} \in \mathbb{R}^n$, we have that $\langle M^\top \mathbf{u}^*, -\mathbf{x} \rangle = -\langle M^\top \mathbf{u}^*, \mathbf{x} \rangle \leq 0$. This then implies that $\langle M^\top \mathbf{u}^*, \mathbf{x} \rangle = 0$, as it is the only number that equals its own negation. By the positive-semidefiniteness of inner products, it follows that

$$M^\top \mathbf{u}^* = 0. \quad \square$$

Observe that, we want only the solution where $M\mathbf{x} \leq \mathbf{b}$ in (P0), which is equivalent to $M\mathbf{x} - \mathbf{b} \in \mathbb{R}_-^m$ and, otherwise, discard may as $-\infty$. Consider the following notion to reformulate the primal problem (P0) in a more analytical form. The *geometrical indicator function* is denoted and defined as $\delta_S : \mathbb{R}^m \rightarrow \mathbb{R} \cup \{\infty\}$ where

$$\delta_S(\mathbf{s}) = \begin{cases} 0 & \text{if } \mathbf{s} \in S, \text{ and} \\ \infty & \text{otherwise.} \end{cases}$$

The subscript is sometimes replaced by an event with respect to the input; for example, if $S = \{\mathbf{x} \in \mathbb{R}^n : M\mathbf{x} \leq \mathbf{b}\}$ for some $M \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$, then we may write $\delta_S(\mathbf{x}_0) = \delta_{M\mathbf{x}_0 \leq \mathbf{b}}(\mathbf{x}_0)$. We now reformulate the program (P0) as

$$z^* = \max\{\langle \mathbf{c}, \mathbf{x} \rangle : M\mathbf{x} \leq \mathbf{b}, \mathbf{x} \in \mathbb{R}^n\} = \sup_{\mathbf{x} \in \mathbb{R}^n} \left\{ \langle \mathbf{c}, \mathbf{x} \rangle - \delta_{\mathbb{R}_-^m}(M\mathbf{x} - \mathbf{b}) \right\} \quad (2.5) \quad \square \text{a}$$

where $z^* \in \mathbb{R} \cup \{-\infty\}$. Recall that, given a cone $S \subseteq V$, the polar cone is denoted and defined as $S^\circ = \{\mathbf{y} \in V : \forall \mathbf{x} \in S, \langle \mathbf{x}, \mathbf{y} \rangle \leq 0\}$.

1 representation)

Proposition 2.6 (Support function Representation via Fenchel-Moreau). *Suppose that S is a closed convex cone. We have that $\delta_S(\mathbf{x}) = \sup\{\langle \mathbf{x}, \mathbf{u} \rangle : \mathbf{u} \in S^\circ\}$. $\square$*

Proposition 2.6 then rewrites the program defined in (2.5) as

$$\begin{aligned} z^* &= \sup_{\mathbf{x} \in \mathbb{R}^n} \left\{ \langle \mathbf{c}, \mathbf{x} \rangle - \sup_{\mathbf{u} \in \mathbb{R}_+^m} \{\langle \mathbf{u}, M\mathbf{x} - \mathbf{b} \rangle\} \right\} \\ &= \sup_{\mathbf{x} \in \mathbb{R}^n} \left\{ \langle \mathbf{c}, \mathbf{x} \rangle + \inf_{\mathbf{u} \in \mathbb{R}_+^m} \{\langle \mathbf{u}, \mathbf{b} \rangle - \langle \mathbf{u}, M\mathbf{x} \rangle\} \right\}, \end{aligned}$$

which finally gets to

$$z^* = \sup_{\mathbf{x} \in \mathbb{R}^n} \inf_{\mathbf{u} \in \mathbb{R}_+^m} \left\{ \langle \mathbf{c}, \mathbf{x} \rangle + \langle \mathbf{u}, \mathbf{b} \rangle - \langle \mathbf{u}, M\mathbf{x} \rangle \right\}. \quad (2.7) \quad \text{?eqn: } z \sup \inf$$

Given $\mathbf{b} \in \mathbb{R}^m$ and $\mathbf{c} \in \mathbb{R}^n$, we may conveniently define bivariate function L over a subset of $\mathbb{R}^n \times \mathbb{R}^m$ where $L(\mathbf{x}, \mathbf{u}) = \langle \mathbf{c}, \mathbf{x} \rangle + \langle \mathbf{u}, \mathbf{b} \rangle - \langle \mathbf{u}, M\mathbf{x} \rangle$. In the aforementioned definition for z^* , its domain is $\mathbb{R}^n \times \mathbb{R}_+^m$, which then gives $z^* = \sup_{\mathbf{x} \in \mathbb{R}^n} \inf_{\mathbf{u} \in \mathbb{R}_+^m} L(\mathbf{x}, \mathbf{u})$. Now, we consider the following sup-inf exchange. In particular, the following theorem is the fundamental theorem that establishes the connection between the solution of a linear optimization problem and that of its dual.

(minimax)

Theorem 2.8. Let $M \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{c} \in \mathbb{R}^n$. Suppose that L is a bivariate functional defined on $\mathbb{R}^n \times \mathbb{R}_+^m$ such that $L(\mathbf{x}, \mathbf{u}) = \langle \mathbf{c}, \mathbf{x} \rangle + \langle \mathbf{u}, \mathbf{b} \rangle - \langle \mathbf{u}, M\mathbf{x} \rangle$. If there is at least one feasible $\mathbf{x}^*$ for $M\mathbf{x}^* \leq \mathbf{b}$ and the supremum $z^* := \sup_{\mathbf{x} \in \mathbb{R}^n} \inf_{\mathbf{u} \in \mathbb{R}_+^m} L(\mathbf{x}, \mathbf{u})$ is finite, then

$$\sup_{\mathbf{x} \in \mathbb{R}^n} \inf_{\mathbf{u} \in \mathbb{R}_+^m} L(\mathbf{x}, \mathbf{u}) = \inf_{\mathbf{u} \in \mathbb{R}_+^m} \sup_{\mathbf{x} \in \mathbb{R}^n} L(\mathbf{x}, \mathbf{u}). \quad (2.9) \quad \text{eqn: sup inf exc}$$

Proof: Let $X = \mathbb{R}^n$ and $U = \mathbb{R}_+^m$. We first prove that

$$\sup_{\mathbf{x} \in X} \inf_{\mathbf{u} \in U} L(\mathbf{x}, \mathbf{u}) \leq \inf_{\mathbf{u} \in U} \sup_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{u}),$$

purely by the definitions of supremum and infimum. For all $\mathbf{x}_0 \in X$ and $\mathbf{u}_0 \in U$, the following inequality holds by definition

$$\inf_{\mathbf{u} \in U} L(\mathbf{x}_0, \mathbf{u}) \leq L(\mathbf{x}_0, \mathbf{u}_0) \leq \sup_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{u}_0).$$

Following precisely from the definition of supremum, the condition that for all $\mathbf{x}_0 \in X$, $\inf_{\mathbf{u} \in U} L(\mathbf{x}_0, \mathbf{u}) \leq \sup_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{u}_0)$ provides the following inequality

$$\sup_{\mathbf{x} \in X} \inf_{\mathbf{u} \in U} L(\mathbf{x}, \mathbf{u}) \leq \sup_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{u}_0).$$

Again, since $\mathbf{u}_0 \in U$ is chosen arbitrarily, we have that

$$\sup_{\mathbf{x} \in X} \inf_{\mathbf{u} \in U} L(\mathbf{x}, \mathbf{u}) \leq \inf_{\mathbf{u} \in U} \sup_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{u}),$$

which proves one direction.

We now turn to the other direction that $\sup_{\mathbf{x} \in X} \inf_{\mathbf{u} \in U} L(\mathbf{x}, \mathbf{u}) \geq \inf_{\mathbf{u} \in U} \sup_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{u})$; toward a contradiction, suppose that $\sup_{\mathbf{x} \in X} \inf_{\mathbf{u} \in U} L(\mathbf{x}, \mathbf{u}) < \inf_{\mathbf{u} \in U} \sup_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{u})$. Observe that the polar dual of $\mathbb{R}^n$ is $(\mathbb{R}^n)^\circ = \{0\}$. We may apply the support function representation so that, for a fixed $\mathbf{u}_0$, we have

$$\begin{aligned} \sup_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{u}_0) &= \sup_{\mathbf{x} \in X} \{ \langle \mathbf{c}, \mathbf{x} \rangle + \langle \mathbf{u}_0, \mathbf{b} \rangle - \langle \mathbf{u}_0, M\mathbf{x} \rangle \} \\ &= \langle \mathbf{u}_0, \mathbf{b} \rangle + \sup_{\mathbf{x} \in \mathbb{R}^n} \{ \langle \mathbf{c}, \mathbf{x} \rangle - \langle \mathbf{u}_0, M\mathbf{x} \rangle \} \\ &= \langle \mathbf{u}_0, \mathbf{b} \rangle + \sup_{\mathbf{x} \in \mathbb{R}^n} \{ \langle \mathbf{c} - M^\top \mathbf{u}_0, \mathbf{x} \rangle \} \\ &= \langle \mathbf{u}_0, \mathbf{b} \rangle + \delta_{\{\mathbf{u}: \mathbf{c} - M^\top \mathbf{u} = 0\}}(\mathbf{u}_0). \end{aligned}$$

By the contradiction hypothesis and the definition of an infimum, fix any $\mathbf{u}_0 \in \mathbb{R}_+^m$, we

have the following chain of inequalities that

$$\sup_{\mathbf{x} \in \mathbb{R}^n} \inf_{\mathbf{u} \in \mathbb{R}_+^m} L(\mathbf{x}, \mathbf{u}) < \inf_{\mathbf{u} \in \mathbb{R}_+^m} \sup_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{u}) \leq \langle \mathbf{u}_0, \mathbf{b} \rangle + \delta_{\{\mathbf{u}: M^\top \mathbf{u} = \mathbf{c}\}}(\mathbf{u}_0),$$

and $\langle \mathbf{u}_0, \mathbf{b} \rangle + \delta_{\{\mathbf{u}: M^\top \mathbf{u} = \mathbf{c}\}}(\mathbf{u}_0) \in \{\langle \mathbf{u}_0, \mathbf{b} \rangle, \infty\}$. Thus, as we verified the two sides of the strict inequality and by the density of the real numbers, we are ensured that there is an $\alpha \in \mathbb{R}$ so that

$$\max_{\mathbf{x} \in \mathbb{R}^n} \{\langle \mathbf{c}, \mathbf{x} \rangle : M\mathbf{x} \leq \mathbf{b}\} =: z^* < \alpha < \inf_{\mathbf{u} \in \mathbb{R}_+^m} \sup_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{u}).$$

Since z^* is the supremum, that is, $\langle \mathbf{c}, \mathbf{x}_0 \rangle < \alpha$ for all $\mathbf{x}_0 \in \mathbb{R}^n$, the following system does not have a solution

$$\begin{bmatrix} M \\ -\mathbf{c}^\top \end{bmatrix} \mathbf{x} \leq \begin{bmatrix} \mathbf{b} \\ -\alpha \end{bmatrix}.$$

By Theorem 2.1, let $\mathbf{u} = (u_1, u_2, \dots, u_m)^\top$ and u_c be variables, the following systems of inequalities

$$\begin{aligned} [M^\top \mid -\mathbf{c}] \begin{bmatrix} \mathbf{u} \\ u_c \end{bmatrix} &= \mathbf{0} \\ \begin{bmatrix} \mathbf{u} \\ u_c \end{bmatrix} &\geq \mathbf{0} \\ \left\langle \begin{bmatrix} \mathbf{u} \\ u_c \end{bmatrix}, \begin{bmatrix} \mathbf{b} \\ -\alpha \end{bmatrix} \right\rangle &< 0 \end{aligned} \tag{2.10} \text{ inhom}$$

has a solution. Let $\mathbf{u}^*$ and u_c^* be a solution for the system (2.10), and reformulate so we have

$$M^\top \mathbf{u}^* = u_c^* \mathbf{c}, \quad \mathbf{u}^* \geq \mathbf{0}, \quad u_c^* \geq 0, \quad \langle \mathbf{u}^*, \mathbf{b} \rangle < \alpha u_c^*. \tag{2.11} \text{ eqn: aux D}$$

We first assume $u_c^* \neq 0$, which (2.11) gives that $u_c^* > 0$. Let $\tilde{\mathbf{u}} = \mathbf{u}^*/u_c^*$; since $u_c^* \geq 0$, we obtain that

$$M^\top \tilde{\mathbf{u}} = \mathbf{c}, \quad \tilde{\mathbf{u}} = \frac{\mathbf{u}^*}{u_c^*} \geq \mathbf{0}, \quad \langle \tilde{\mathbf{u}}, \mathbf{b} \rangle = \frac{1}{u_c^*} \langle \mathbf{u}^*, \mathbf{b} \rangle < \alpha.$$

However, we now have $\delta_{M^\top \tilde{\mathbf{u}} = \mathbf{c}}(\tilde{\mathbf{u}}) = 0$, which leads to the contradiction

$$\langle \tilde{\mathbf{u}}, \mathbf{b} \rangle < \alpha < \inf_u \sup_x L(x, u) \leq \langle \tilde{\mathbf{u}}, \mathbf{b} \rangle,$$

where α is assumed to be strictly less than the infimum and the infimum is the lower bound for $\langle \tilde{\mathbf{w}}, \mathbf{b} \rangle$ by definition.

Alternatively, we assume $u_c^* = 0$ and plug in to inequalities (2.11) and

$$M^\top \mathbf{u}^* = \mathbf{0}, \quad \mathbf{u}^* \geq \mathbf{0}, \quad \langle \mathbf{u}^*, \mathbf{b} \rangle < 0.$$

As we assumed its existence, let $\mathbf{x}_0$ be a feasible solution to the system of inequalities $M\mathbf{x} \leq \mathbf{b}$. Since $M^\top \mathbf{u}^* = \mathbf{0}$ and $\mathbf{u} \geq \mathbf{0}$, we now have a contradiction that

$$0 = \langle M^\top \mathbf{u}^*, \mathbf{x}_0 \rangle = \langle \mathbf{u}^*, M\mathbf{x}_0 \rangle \leq \langle \mathbf{u}^*, \mathbf{b} \rangle < 0.$$

This completes the proof. $\square$

We observe from the proof of Theorem 2.1 that, the change of the constraints and bounds in the primal problem (2.2) leads to the change in the constraints in its dual problem (2.3).

- In (2.3), the constraint $A^\top \mathbf{y} = \mathbf{0}$ is determined by the fact that $\mathbf{x}$ is taken from the entire space of $\mathbb{R}^n$. If we instead restrict $\mathbf{x} \in \mathbb{R}_+^n$ and change the proof accordingly, then the dual constraints in (2.3) change to $A^\top \mathbf{y} \geq \mathbf{0}$. Indeed, in the proof, when the slack variable satisfies $\mathbf{s} = \mathbf{0}$, then $\langle \mathbf{y}^*, A\mathbf{x} \rangle = \langle A^\top \mathbf{y}^*, \mathbf{x} \rangle \geq 0$ for all $\mathbf{x} \geq \mathbf{0}$ if and only if $A^\top \mathbf{y} \geq \mathbf{0}$. More generally, for any closed convex cone C , if $\mathbf{x} \in C$, then the dual condition becomes

$$M^\top \mathbf{u} \in C^* \text{ where } C^* := \{\mathbf{y} \in \mathbb{R}^n : \forall \mathbf{x} \in C, \langle \mathbf{x}, \mathbf{y} \rangle \geq 0\}.$$

where C^* is known as the *dual cone* of C .

- The non-negativeness of $\mathbf{u} \in \mathbb{R}_+^m$ is independent of the primal cone C of where $\mathbf{x}$ is chosen. But dictated by the inequality direction $M\mathbf{x} \leq \mathbf{b}$. Observe that if we instead have $M\mathbf{x} \geq \mathbf{b}$, then we would take $Z = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} + \mathbb{R}_-^m$, which then gives $\mathbf{y} \leq \mathbf{0}$.
- In a similar vein, the bivariate Lagrangian function in Theorem 2.8 can be defined more generally.

?<conic minimax>?

Theorem 2.12 (Conic Minimax Theorem). *Let $M \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{c} \in \mathbb{R}^n$. Let $X \subseteq \mathbb{R}^n$ and $U \subseteq \mathbb{R}^m$ be closed convex cones and the bivariate Lagrangian $L : X \times U \rightarrow \mathbb{R}$ is defined as $L(\mathbf{x}, \mathbf{u}) = \langle \mathbf{c}, \mathbf{x} \rangle + \langle \mathbf{u}, \mathbf{b} \rangle - \langle \mathbf{u}, M\mathbf{x} \rangle$. If there is at least one feasible $\mathbf{x}^* \in X$ so that $\mathbf{b} - M\mathbf{x}^* \in U$ and the supremum $z^* := \sup_{\mathbf{x} \in X} \inf_{\mathbf{u} \in U^*} L(\mathbf{x}, \mathbf{u})$ is finite, then*

$$\sup_{\mathbf{x} \in X} \inf_{\mathbf{u} \in U^*} L(\mathbf{x}, \mathbf{u}) = \inf_{\mathbf{u} \in U^*} \sup_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{u}). \quad \square$$

Finally, note that the system (2.10) is precisely how to transform an optimization problem with an inhomogeneous system into a homogeneous system with no optimization

tasks. As such, and along with the remarks above, we have the formal and complete form of a linear program as follows. Let $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, and $\mathbf{c} \in \mathbb{R}^n$; let $L \subseteq \mathbb{R}^m$ and $K \subseteq \mathbb{R}^n$ be two closed convex cones. Let $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^m$ be two vectors of variables. When we define the Linear Problem (GP) as

$$\begin{array}{ll} \max_{\mathbf{x}} & \langle \mathbf{c}, \mathbf{x} \rangle \\ \text{subject to} & \mathbf{b} - A\mathbf{x} \in L, \\ & \mathbf{x} \in K; \end{array} \quad (\text{GP}) \quad \boxed{\text{GP}}$$

the *dual* problem of (GP) follows and defines the Linear Problem (GD) as

$$\begin{array}{ll} \min_{\mathbf{y}} & \langle \mathbf{y}, \mathbf{b} \rangle \\ \text{subject to} & A^\top \mathbf{y} - \mathbf{c} \in K^*, \\ & \mathbf{y} \in L^*. \end{array} \quad (\text{GD}) \quad \boxed{\text{GD}}$$

If we choose $L = \mathbb{R}_+^m$ and $K = \mathbb{R}_+^n$, then we can define a primal problem (P) such that

$$\begin{array}{ll} \max_{\mathbf{x} \in \mathbb{R}_+^n} & \langle \mathbf{c}, \mathbf{x} \rangle \\ \text{subject to} & A\mathbf{x} \leq \mathbf{b}, \\ & \mathbf{x} \geq 0, \end{array} \quad (\text{P}) \quad \boxed{\text{P}}$$

and its dual problem (D) as

$$\begin{array}{ll} \min_{\mathbf{y} \in \mathbb{R}_+^m} & \langle \mathbf{y}, \mathbf{b} \rangle \\ \text{subject to} & A^\top \mathbf{y} \geq \mathbf{c}, \\ & \mathbf{y} \geq 0. \end{array} \quad (\text{D}) \quad \boxed{\text{D}}$$

In particular, (P) has a solution if and only if (D) has a solution and, if (P) has a solution, then the maximum it attains corresponds to a solution to (D) that attains the same value as the minimum. However, there is a caveat: even if we are given an optimal solution to a primal problem, we do not immediately obtain an optimal solution for its dual problem, as the strong duality only guarantees the existence of an optimal solution, not an explicit construction. A relevant constructive solution will be discussed in Chapter 4.

2.2 Geometric Point of View for Dual Problems

In Theorem 2.1, we considered the existential argument for the feasibility of a system $M\mathbf{x} \leq \mathbf{b}$. We now turn to a constructive description of Farkas' Lemma. In particular, we state it in a less general form when we also consider the equality constraints; yet, an equality constraint $ax = b$ can be expressed as two inequalities $ax \leq b$ and $-ax \leq b$ and, thus, equality constraints are not considered special cases for systems of linear inequalities.

$\langle \text{farkas geo} \rangle$

Theorem 2.13 (Geometric Farkas' Lemma). *Let m, d_E, d_A be positive integers and $E \in \mathbb{R}^{m \times d_E}, A \in \mathbb{R}^{m \times d_A}$, and $Y_{E,A} = \{\mathbf{y} \in \mathbb{R}^m : E^\top \mathbf{y} = \mathbf{0}, A^\top \mathbf{y} \leq \mathbf{0}\}$. A vector $\mathbf{b} \in \mathbb{R}^m$ satisfies $\langle \mathbf{y}, \mathbf{b} \rangle \leq 0$ for every $\mathbf{y} \in Y_{E,A}$ if and only if there exists $\boldsymbol{\lambda} \in \mathbb{R}^{d_E}$ and $\boldsymbol{\mu} \in \mathbb{R}_+^{d_A}$ so that*

$$\mathbf{b} = E\boldsymbol{\lambda} + A\boldsymbol{\mu}. \quad (2.14) \quad \text{eqn: farkas geo}$$

Proof: First, we reformulate the constraint (2.14) into the following system for $\boldsymbol{\lambda} \in \mathbb{R}^{d_E}$ and $\boldsymbol{\mu} \in \mathbb{R}^{d_A}$ such that

$$\begin{bmatrix} E & A \\ -E & -A \\ \mathbf{O} & -I_{d_A} \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ \boldsymbol{\mu} \end{bmatrix} \leq \begin{bmatrix} \mathbf{b} \\ -\mathbf{b} \\ \mathbf{0} \end{bmatrix}, \quad (2.15) \quad \text{eqn: farkas geo}$$

where $\mathbf{O}$ denotes an all-zero matrix with appropriate dimensions. The two blocks $E\boldsymbol{\lambda} + A\boldsymbol{\mu} \leq \mathbf{b}$ and $-E\boldsymbol{\lambda} - A\boldsymbol{\mu} \leq -\mathbf{b}$ corresponds to the equality, and the block $\mathbf{O}\boldsymbol{\lambda} - I\boldsymbol{\mu} \leq \mathbf{0}$ corresponds to that $\boldsymbol{\mu} \geq \mathbf{0}$. Since we will apply Theorem 2.1, consider the following dual system of equalities and inequalities with a list of variables $\mathbf{u} \in \mathbb{R}^{m+m+d_A}$:

$$\begin{aligned} \begin{bmatrix} E^\top & -E^\top & \mathbf{O} \\ A^\top & -A^\top & -I_{d_A} \end{bmatrix} \mathbf{u} &= \mathbf{0}, \\ \mathbf{u} &\geq \mathbf{0}, \\ \left\langle \mathbf{u}, \begin{bmatrix} \mathbf{b} \\ -\mathbf{b} \\ \mathbf{0} \end{bmatrix} \right\rangle &< 0 \end{aligned} \quad (2.16) \quad \text{eqn: geo farkas}$$

We first prove the forward implication; suppose $\langle \mathbf{y}, \mathbf{b} \rangle \leq 0$ for every $\mathbf{y} \in Y_{E,A}$ and, toward a contradiction, that the system (2.15) has no solution; by Theorem 2.1, there is a solution $\tilde{\mathbf{u}} \in \mathbb{R}^{m+m+d_A}$ for (2.16). We partition the vector

$$\tilde{\mathbf{u}} = \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \mathbf{y}_3 \end{bmatrix},$$

where $\mathbf{y}_1, \mathbf{y}_2 \in \mathbb{R}_+^m$ and $\mathbf{y}_3 \in \mathbb{R}_+^{d_A}$ and the non-negativeness is given that that $\tilde{\mathbf{u}} \geq \mathbf{0}$, so that $E^\top \mathbf{y}_1 - E^\top \mathbf{y}_2 = \mathbf{0}$ and $A^\top \mathbf{y}_1 - A^\top \mathbf{y}_2 - \mathbf{y}_3 = \mathbf{0}$. Equivalently, we write that

$$E^\top(\mathbf{y}_1 - \mathbf{y}_2) = \mathbf{0}, \quad A^\top(\mathbf{y}_1 - \mathbf{y}_2) = \mathbf{y}_3 = \mathbf{0}, \quad \text{and } \langle \mathbf{y}_1, \mathbf{b} \rangle - \langle \mathbf{y}_2, \mathbf{b} \rangle < 0,$$

where the last inequality is due to $\langle \mathbf{y}_1, \mathbf{b} \rangle + \langle \mathbf{y}_2, -\mathbf{b} \rangle + \langle \mathbf{y}_3, \mathbf{0} \rangle < 0$ as in (2.16). Now, we set $\mathbf{y}_0 = \mathbf{y}_2 - \mathbf{y}_1$ and we have that $E^\top \mathbf{y}_0 = \mathbf{0}, A^\top \mathbf{y}_0 \leq \mathbf{0}$; this means $\mathbf{y}_0 \in Y_{E,A}$. Finally, $\langle \mathbf{y}_0, \mathbf{b} \rangle > 0$ contradicts the premise that $\langle \mathbf{y}, \mathbf{b} \rangle \leq 0$ for all $\mathbf{y} \in Y_{E,A}$.

For the converse direction, suppose that $\mathbf{b} = E\boldsymbol{\lambda}_0 + A\boldsymbol{\mu}_0$ for some $\boldsymbol{\lambda}_0 \in \mathbb{R}^{d_E}, \boldsymbol{\mu}_0 \in \mathbb{R}_+^{d_A}$. We assume that $Y_{E,A} \neq \emptyset$ or, otherwise, we are done. By the existence of $\boldsymbol{\lambda}_0$ and $\boldsymbol{\mu}_0$, the

system (2.15) has a solution and, by Theorem 2.1, the system (2.16) has no solution. In the following, we fix a $\mathbf{y} \in Y_{E,A}$ and we will show that $\langle \mathbf{y}, \mathbf{b} \rangle \leq 0$.

For $i \in \{1, 2, \dots, m\}$, let y_i denote the i -th component of $\mathbf{y}$; we define $\mathbf{u}^+, \mathbf{u}^- \in \mathbb{R}^m$ where each of their i -th component has that

$$u_i^+ = \begin{cases} y_i & \text{if } y_i \geq 0 \\ 0 & \text{otherwise,} \end{cases} \quad \text{and} \quad u_i^- = \begin{cases} -y_i & \text{if } y_i < 0 \\ 0 & \text{otherwise,} \end{cases}$$

so that $\mathbf{y} = \mathbf{u}^+ - \mathbf{u}^-$. We now set

$$\mathbf{u}^* = \begin{bmatrix} \mathbf{u}^- \\ \mathbf{u}^+ \\ -A^\top \mathbf{y} \end{bmatrix} \in \mathbb{R}^{m+m+d_A}.$$

Since $\mathbf{y} \in Y_{E,A}$, we have $\mathbf{0} = E^\top \mathbf{y} = E^\top (\mathbf{u}^+ - \mathbf{u}^-) = E^\top \mathbf{u}^- - E^\top \mathbf{u}^+$; we also have that $\mathbf{u}^* \geq \mathbf{0}$ since $A^\top \mathbf{y} \leq \mathbf{0}$. That says,

$$\left[\begin{array}{c|c|c} E^\top & -E^\top & \mathbf{0} \\ A^\top & -A^\top & -I \end{array} \right] \begin{bmatrix} \mathbf{u}^- \\ \mathbf{u}^+ \\ -A^\top \mathbf{y} \end{bmatrix} = \begin{bmatrix} E^\top \mathbf{u}^- - E^\top \mathbf{u}^+ \\ A^\top \mathbf{u}^- - A^\top \mathbf{u}^+ + A^\top \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} = \mathbf{0}.$$

Since the system (2.16) has no solution, we have that

$$-\langle \mathbf{y}, \mathbf{b} \rangle = \langle -\mathbf{y}, \mathbf{b} \rangle = \langle \mathbf{u}^-, \mathbf{b} \rangle - \langle \mathbf{u}^+, \mathbf{b} \rangle = \left\langle \begin{bmatrix} \mathbf{u}^- \\ \mathbf{u}^+ \\ -A^\top \mathbf{y} \end{bmatrix}, \begin{bmatrix} \mathbf{b} \\ -\mathbf{b} \\ \mathbf{0} \end{bmatrix} \right\rangle \geq 0$$

and, thus $\langle \mathbf{y}, \mathbf{b} \rangle \leq 0$. □

We now consider the geometric intuition behind Theorem 2.13. Consider the matrices and their columns

$$E = [\mathbf{e}_1 \mid \mathbf{e}_2 \mid \dots \mid \mathbf{e}_{d_E}] \quad \text{and} \quad A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \dots \mid \mathbf{a}_{d_A}];$$

the theorem states that the necessary condition, Equation 2.14

$$\mathbf{b} \in \text{span}(\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{d_E}\}) \cup \text{cone}(\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{d_A}\})$$

happens if and only if, for every vector $\mathbf{y}$ that is orthogonal to every $\mathbf{e}_i$ and oppose to every $\mathbf{a}_i$, $\mathbf{y}$ is also oppose to $\mathbf{b}$.

Chapter 3

Polyhedral Geometry

polyhedral geometry In this section, we take motivation from Farkas' Lemma and its geometric version, defining optimization in terms of a broader geometric concept on cones and observing equivalences from a higher and more abstract point of view. A set $S \subseteq \mathbb{R}^m$ is *polyhedral* if there is a finite list of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^m$ and a list of bounds $b_1, b_2, \dots, b_n \in \mathbb{R}$ so that

$$S = \{\mathbf{x} \in \mathbb{R}^m : \forall i \in \{1, 2, \dots, n\}, \langle \mathbf{x}, \mathbf{v}_i \rangle \leq b_i\}$$

or equivalently, transpose and stack all $\mathbf{v}_i$ and there is a finite-dimensional constraint matrix $M \in \mathbb{R}^{n \times m}$, and $\mathbf{b} = (b_1, b_2, \dots, b_n)^\top$ so that

$$S = \{\mathbf{x} \in \mathbb{R}^m : M\mathbf{x} \leq \mathbf{b}\}.$$

One should have observed that this is essentially the feasible region mentioned for a primal problem (P). When we have a homogeneous system, where the bound $\mathbf{b} = \mathbf{0}$, a polyhedral set is a *polyhedral cone*. A *linear half-space*, or just a *halfspace*, defined by $\mathbf{a} \in \mathbb{R}^m$ is the set

$$H_{\mathbf{a}} = \{\mathbf{y} \in \mathbb{R}^m : \langle \mathbf{a}, \mathbf{y} \rangle \leq 0\}.$$

We take the fact that every halfspace is closed and convex as given. A polyhedral cone can be understood as the intersection of finitely many half-spaces

$$H_{\mathbf{v}_1} \cap H_{\mathbf{v}_2} \cap \dots \cap H_{\mathbf{v}_n}.$$

closed and convex **Proposition 3.1.** *Every polyhedral set is closed and convex. In particular, a polyhedral set P has that $P = (P^\circ)^\circ$.*

Proof: Consider the definition of a polyhedral set as a finite intersection of closed half-spaces. By Proposition 1.1 and 1.4, we have that every polyhedral set remains closed and convex. $\square$

Remark 3.2. With the definition of a polyhedral cone, the geometric version of Farkas' Lemma, Theorem 2.13 can be reinterpreted as follows. If a vector $\mathbf{b}$ is in the polar of a polyhedral cone defined by $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r$, then $\mathbf{b} \in \text{cone}(\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r\})$.

3.1 Structure and Representations of Polyhedra

In plain language, the Minkowski-Weyl Theorem is very direct: a cone is finitely generated (a vertex representation, such as the geometric version of Farkas' Lemma) if and only if it can be represented by a finite list of hyperplane boundaries (a hyperplane representation, such as Theorem 2.1). One may naturally expect that there is a very short, easy proof via some auxiliary matrices as in the proofs of Theorem 2.1, 2.8, or 2.13; unfortunately, no, it still requires a last piece of sophisticated, yet elegant machinery and result to finally prove the Minkowski-Weyl Theorem.

3.1.1 The Fourier-Motzkin Elimination Algorithm Framework

We sketch the proof for the Minkowski-Weyl Theorem briefly. Let $C = \text{cone}(\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\})$ be finitely generated cone; we would show that C is also a polyhedral cone in $\mathbb{R}^m$ with respect to some matrix M with m columns. Since every $\mathbf{b} \in C$ has a representation of $\mathbf{b} = \mu_1 \mathbf{a}_1 + \mu_2 \mathbf{a}_2 + \dots + \mu_n \mathbf{a}_n$. Let $A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \dots \mid \mathbf{a}_n]$ and define matrix

$$M_{m+n} = \begin{bmatrix} I_m & -A \\ -I_m & A \\ \mathbf{0}_{n \times m} & -I_n \end{bmatrix}$$

so that every concatenation $\mathbf{x} = (\mathbf{b}, \boldsymbol{\mu}) \in \mathbb{R}^{m+n}$, for $\mathbf{b} \in \mathbb{R}^m$ and the corresponding $\boldsymbol{\mu} \in \mathbb{R}_+^n$ where $\mathbf{b} = A\boldsymbol{\mu}$, is a solution to the system of inequalities $M_{m+n}\mathbf{x} \leq \mathbf{0}$.

Now, the matrix M_{m+n} defines a polyhedral cone

$$P = \{\mathbf{x} \in \mathbb{R}^{m+n} : M_{m+n}\mathbf{x} \leq \mathbf{0}\}$$

in which one may verify that $P = \{(\mathbf{b}, \boldsymbol{\mu}) \in \mathbb{R}^{m+n} : \boldsymbol{\mu} \in \mathbb{R}_+^n, \mathbf{b} = A\boldsymbol{\mu}\}$. We would then need to show that the polyhedral cone $P \subseteq \mathbb{R}^{m+n}$ can be projected down to some set $P' \subseteq \mathbb{R}^m$ and P' is a polyhedral cone defined by some M_m so that $M_m\mathbf{b} \leq \mathbf{0}$ if and only if $\mathbf{b} \in P'$.

To obtain the matrix M_m , we will perform the *Fourier-Motzkin elimination* algorithm. The algorithm projects a polyhedral cone $C_{k+1} \subseteq \mathbb{R}^{k+1}$ to some $C_k \subseteq \mathbb{R}^k$ that, first, remains a polyhedral cone and, second, every $\mathbf{x} \in C_k$ is a part of a solution in C_{k+1} : formally, we state the Fourier-Motzkin Elimination framework below.

Theorem 3.3 (Fourier-Motzkin Elimination). *Let m and n be positive integers. Suppose $P_{n+1} = \{\boldsymbol{\mu} \in \mathbb{R}^{n+1} : A\boldsymbol{\mu} \leq \mathbf{b}\} \subseteq \mathbb{R}^{n+1}$ is a polyhedral set defined by a matrix $A \in \mathbb{R}^{m \times (n+1)}$ and $\mathbf{b} \in \mathbb{R}^m$, there then are $\mathbf{a}'_1, \mathbf{a}'_2, \dots, \mathbf{a}'_{m'} \in \mathbb{R}^n$ and $b'_1, b'_2, \dots, b'_{m'} \in \mathbb{R}$ that defines a polyhedral set $P_n = \{\mathbf{x} \in \mathbb{R}^n : \forall i \in \{1, 2, \dots, m\}, \langle \mathbf{a}_i, \mathbf{x} \rangle \leq b'_i\} \subseteq \mathbb{R}^n$ with the projection property*

$$\mathbf{x} \in P_n \iff \exists x_{n+1} \in \mathbb{R} \text{ such that } \begin{bmatrix} \mathbf{x} \\ x_{n+1} \end{bmatrix} \in P_{n+1}.$$

Proof: Suppose that $A \in \mathbb{R}^{m \times (n+1)}$, $\mathbf{b} \in \mathbb{R}^m$, $\boldsymbol{\mu} \in \mathbb{R}^{n+1}$ are as in the statement so that $A\boldsymbol{\mu} \leq \mathbf{b}$ defines the polyhedral set P ; let $\mathbf{x}$ be the first n components of $\boldsymbol{\mu}$ and x_{n+1} be the last component; i.e., $\boldsymbol{\mu} = (\mathbf{x}, x_{n+1}) \in \mathbb{R}^{n+1}$. For $i \in \{1, 2, \dots, m\}$ and $j \in \{1, 2, \dots, n+1\}$, let $a_{i,j} \in \mathbb{R}$ denote the (i, j) -th entry of A and b_i be the i -th entry of $\mathbf{b}$. For $i \in \{1, 2, \dots, m\}$, we consider take first n entries if the i -th row and denote them as $\mathbf{a}_i^\top = [a_{i,1}, a_{i,2}, \dots, a_{i,n}]$ so that we have $\langle \mathbf{a}_i, \mathbf{x} \rangle + a_{i,n+1}x_{n+1} \leq b_i$, where b_i is the i -th entry of $\mathbf{b}$. In particular, we now have the format

$$\left[\begin{array}{ccc|c} \text{---} & \mathbf{a}_1^\top & \text{---} & a_{1,n+1} \\ \text{---} & \mathbf{a}_2^\top & \text{---} & a_{2,n+1} \\ & \vdots & & \vdots \\ \text{---} & \mathbf{a}_m^\top & \text{---} & a_{m,n+1} \end{array} \right] \left[\begin{array}{c} | \\ \mathbf{x} \\ | \\ x_{n+1} \end{array} \right] \leq \left[\begin{array}{c} b_1 \\ b_2 \\ \vdots \\ b_m \end{array} \right].$$

First, we partition the index set $\{1, 2, \dots, m\} = J_0 \cup J_+ \cup J_-$ where

$$\begin{aligned} J_0 &= \{i : a_{i,n+1} = 0\}, \\ J_+ &= \{i : a_{i,n+1} > 0\}, \text{ and} \\ J_- &= \{i : a_{i,n+1} < 0\}. \end{aligned}$$

Thus, when $i \in J_0$, we have $\langle \mathbf{a}_i, \mathbf{x} \rangle \leq b_i$ remain the same; for $i \in J_+$, since $a_{i,n+1} > 0$, we have

$$x_{n+1} \leq \frac{b_i - \langle \mathbf{a}_i, \mathbf{x} \rangle}{a_{i,n+1}};$$

for $i \in J_-$, the inequality is flipped by dividing a negative number $a_{i,n+1} < 0$ so we have

$$x_{n+1} \geq \frac{b_i - \langle \mathbf{a}_i, \mathbf{x} \rangle}{a_{i,n+1}}.$$

Next, for each pair $(k, \ell) \in J_- \times J_+$, since we have $\boldsymbol{\mu} = (\mathbf{x}, x_{n+1})$ is a solution for $A\boldsymbol{\mu} \leq \mathbf{b}$, we have that

$$\frac{b_k - \langle \mathbf{a}_k, \mathbf{x} \rangle}{a_{k,n+1}} \leq \frac{b_\ell - \langle \mathbf{a}_\ell, \mathbf{x} \rangle}{a_{\ell,n+1}}. \quad (3.4) \quad \text{eqn: FW squeeze}$$

We multiply both sides by $-a_{k,n+1}a_{\ell,n+1}$, which is positive since $k \in J_-$ and $-a_{k,n+1} > 0$;

this eliminates the denominator while keeping the direction of inequalities. Observe the following sequence of equivalence

$$\begin{aligned}
& -a_{k,n+1}a_{\ell,n+1} \frac{b_k - \langle \mathbf{a}_k, \mathbf{x} \rangle}{a_{k,n+1}} \leq -a_{k,n+1}a_{\ell,n+1} \frac{b_\ell - \langle \mathbf{a}_\ell, \mathbf{x} \rangle}{a_{\ell,n+1}} \\
\iff & -a_{\ell,n+1}(b_k - \langle \mathbf{a}_k, \mathbf{x} \rangle) \leq -a_{k,n+1}(b_\ell - \langle \mathbf{a}_\ell, \mathbf{x} \rangle) \\
\iff & -a_{\ell,n+1}b_k - (-a_{\ell,n+1}) \langle \mathbf{a}_k, \mathbf{x} \rangle \leq -a_{k,n+1}b_\ell - (-a_{k,n+1}) \langle \mathbf{a}_\ell, \mathbf{x} \rangle \\
\iff & a_{\ell,n+1} \langle \mathbf{a}_k, \mathbf{x} \rangle - a_{k,n+1} \langle \mathbf{a}_\ell, \mathbf{x} \rangle \leq a_{\ell,n+1}b_k - a_{k,n+1}b_\ell \\
\iff & \langle a_{\ell,n+1}\mathbf{a}_k - a_{k,n+1}\mathbf{a}_\ell, \mathbf{x} \rangle \leq a_{\ell,n+1}b_k - a_{k,n+1}b_\ell.
\end{aligned}$$

Now, observe that $J_0 \cup (J_- \times J_+)$ is a finite set so the set defined by

$$\begin{aligned}
P' = \{ \mathbf{x} \in \mathbb{R}^n : \forall i \in J_0, \langle \mathbf{a}_i, \mathbf{x} \rangle \leq b_i \quad \text{and} \\
\forall (k, \ell) \in J_- \times J_+, \langle a_{\ell,n+1}\mathbf{a}_k - a_{k,n+1}\mathbf{a}_\ell, \mathbf{x} \rangle \leq a_{\ell,n+1}b_k - a_{k,n+1}b_\ell \} \quad (3.5) \quad \text{?eqn: FM poly?}
\end{aligned}$$

is a polyhedral set. Meanwhile, every solution in $\mathbf{x} \in P'$ obeys Inequality 3.4 and, thus, we may pick a number between the two bounds as x_{n+1}^* so that the concatenation $\boldsymbol{\mu} = (\mathbf{x}, x_{n+1}^*) \in P$. The proof is done. $\square$

We remark on Theorem 3.3 and its proof; we assume the notation used in the proof. First, we observe that the resulting polyhedral set P' can be obtained by matrix multiplication. Consider a permutating matrix $U_0 \in \mathbb{R}^{|J_0| \times m}$ by appending m rows. Since J_0 is finite, we iterate over J_0 : for each $i^* \in J_0$, we obtain and append to U_0 a row $\mathbf{u}^\top = (u_1, u_2, \dots, u_m)$ with $u_{i^*} = 1$ and $u_j = 0$ for all $j \in \{1, 2, \dots, m\} \setminus \{i^*\}$. In particular, we have

$$\mathbf{u}^\top A = [\text{--- } \mathbf{a}_{i^*}^\top \text{---} \mid 0] \quad \text{and} \quad \mathbf{u}^\top \mathbf{b} = [b_{i^*}] \quad (3.6) \quad \text{eqn: FW zero I}$$

where the last entry is $a_{i^*,n+1} = 0$ as $i^* \in J_0$ by definition. Similarly, define another matrix $U_1 \in \mathbb{R}^{|J_- \times J_+| \times m}$ that sums rows of A . Given the finiteness of J_+ and J_- , we adopt a canonical way to iterate over $J_- \times J_+$. For each $(k^*, \ell^*) \in J_- \times J_+$, define and append $\mathbf{u}^\top = (u_1, u_2, \dots, u_m)$ to U_1 where $u_{k^*} = a_{\ell^*,n+1}$, $u_{\ell^*} = -a_{k^*,n+1}$, and $u_j = 0$ for all $j \in \{1, 2, \dots, m\} \setminus \{\ell^*, k^*\}$. We have that

$$\begin{aligned}
\mathbf{u}^\top A &= [\text{--- } (a_{\ell^*,n+1}\mathbf{a}_{k^*} - a_{k^*,n+1}\mathbf{a}_{\ell^*})^\top \text{---} \mid 0] \quad \text{and} \\
\mathbf{u}^\top \mathbf{b} &= [a_{\ell^*,n+1}b_{k^*} - a_{k^*,n+1}b_{\ell^*}], \quad (3.7) \quad \text{eqn: FW zero I}
\end{aligned}$$

where the last entry 0 is obtained by $0 = a_{\ell^*,n+1}a_{k^*,n+1} + (-a_{k^*,n+1})a_{\ell^*,n+1}$. Let $m' = |J_0| + |J_- \times J_+|$,

$$U = \begin{bmatrix} U_0 \\ U_1 \end{bmatrix} \in \mathbb{R}^{m' \times m},$$

and $D_{n+1,n} \in \mathbb{R}^{n+1,n}$ be the non-standard identity matrix, where every diagonal entry is 1, and 0 otherwise. Due to the observations in Equation 3.6 and 3.7, the last column of

UA an all-zero column, we have that $UA = UAD_{n+1,n}D_{n,n+1}$. Now, observe that U and $D_{n+1,n}$ contains only positive entries; we have that

$$A\mu \leq \mathbf{b} \iff UA\mu \leq U\mathbf{b} \iff (UAD_{n+1,n}D_{n,n+1})\mu = (UAD_{n+1,n})(D_{n,n+1}\mu) \leq U\mathbf{b}. \quad (3.8) \quad \{?\}$$

Finally, we may apply the Fourier-Motzkin Elimination algorithm k times and iteratively obtain $U^{(1)}, U^{(2)}, \dots, U^{(k)}$, and obtain a final matrix $U = U^{(k)}U^{(k-1)} \dots U_1$. Observe that

$$D_{n+k,n+k-1}D_{n+k-1,n+k-2} \dots D_{n+1,n} = D_{n+k,n},$$

which leads to the following corollary. For positive integers p and q , where $p < q$, let $\pi_{q,p} : \mathbb{R}^q \rightarrow \mathbb{R}^p$ be the *projection function* where

$$\pi_{q,p} \left(\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \\ \vdots \\ x_q \end{bmatrix} \right) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}.$$

For a set $Q \subseteq \mathbb{R}^q$, we will denote the set $\Pi_{q,p}(Q) = P$ be the set where

$$P = \{\mathbf{x} \in \mathbb{R}^p : \exists \mathbf{x}' \in \mathbb{R}^q : \pi_{q,p}(\mathbf{x}') = \mathbf{x}\}.$$

r: FM projection

Corollary 3.9. *For a polyhedral set $Q \subseteq \mathbb{R}^{n+k}$, its projection $\Pi_{n+k,n}(Q)$ is a polyhedral set in $\mathbb{R}^n$.*

We remark on the notion of projection: in many textbooks, projection is defined more abstractly in terms of vector spaces; our definition of projection assumes a basis in a vector space and specifically fixes the order of the coordinates. Here, we adopt this style for application reasons, as we will only be dealing with the Euclidean, multi-dimensional spaces of real numbers.

Finally, observe from the proof of Theorem 3.3 that, if $\mathbf{b} = \mathbf{0}$ initially, then the new bounds produced by the algorithm remain $\{b'_1, b'_2, \dots, b'_{m'}\} = \{0\}$.

ne remains cone

Corollary 3.10. *The projection of a polyhedral cone remains a polyhedral cone.*

3.1.2 The Minkowski-Weyl Theorem

Theorem 3.11 (Weyl's Theorem). *Every finitely generated cone is polyhedral; in particular, if $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r \in \mathbb{R}^m$ defines $C = \text{cone}(\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r\})$, then there exists $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_s \in \mathbb{R}^m$ such that $C = \{\mathbf{y} : \forall i \in \{1, 2, \dots, s\}, \langle \mathbf{b}_i, \mathbf{y} \rangle \leq 0\}$.*

Proof: Suppose that the cone $C = \text{cone}(\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r\})$ is finitely generated. Let $A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \dots \mid \mathbf{a}_r]$ and define the matrix

$$M_{m+n} = \begin{bmatrix} I_m & -A \\ -I_m & A \\ \mathbf{0}_{n \times m} & -I_n \end{bmatrix},$$

which defines a polyhedral cone in $Q = \mathbb{R}^{m+n}$ and contains precisely $(\mathbf{b}, \boldsymbol{\mu}) \in \mathbb{R}^{m+n}$ for all $\mathbf{b} \in C$ such that $\mathbf{b} = A\boldsymbol{\mu}$. By Corollary 3.9, $\Pi_{m+n,m}(Q) = C$; in particular, by Theorem 3.3, there is a list of vectors in $\mathbb{R}^m$ that defines the polyhedral C . $\square$

Theorem 3.12 (Minkowski's Theorem). *Every polyhedral cone is finitely generated.*

Proof: Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r \in \mathbb{R}^m$ be a list of vectors that defines the polyhedral cone $P = \{\mathbf{y} \in \mathbb{R}^m : \forall i \in \{1, 2, \dots, r\}, \langle \mathbf{a}_i, \mathbf{y} \rangle \leq 0\}$. Define a matrix $A \in \mathbb{R}^{m \times r}$ where $A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \dots \mid \mathbf{a}_r]$. Let $Y_{\cdot, A} = \{\mathbf{y} \in \mathbb{R}^m : A^\top \mathbf{y} \leq \mathbf{0}\} = P$, we may apply Theorem 2.13 and consider Remark 3.2 so that $P^\circ = \text{cone}(\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r\})$ that the polar is finitely generated. Next, apply Theorem 3.11 so that the polar P° is also polyhedral defined by $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_s \in \mathbb{R}^m$. This implies that $(P^\circ)^\circ$ is finitely generated by $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_s$. Finally, by Proposition 3.1 and the Bipolar Theorem (Theorem 1.9), we have that $P = (P^\circ)^\circ$ and, thus, P is finitely generated. $\square$

Corollary 3.13. *Finitely generated cones are closed and convex.* $\square$

Recall that $\mu_1 \mathbf{v}_1 + \mu_2 \mathbf{v}_2 + \dots + \mu_m \mathbf{v}_m$ is a convex combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ if $\mu_i \in \mathbb{R}_+$ for $i \in \{1, 2, \dots, m\}$ and $\sum_{i=1}^m \mu_i = 1$. The convex hull of a set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ is $\text{conv}(\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\})$ is the set of all convex combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$.

Theorem 3.14 (Minkowski-Weyl Representation). *A set P is polyhedral if and only if there is a set $V = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ and a finitely generated cone C so that $P = \text{conv}(V) + C$.*

Proof: By the definition of a polyhedral set, there is a list $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r \in \mathbb{R}^m$ and $b_1, b_2, \dots, b_r \in \mathbb{R}$ so that $P = \{\mathbf{y} \in \mathbb{R}^m : \forall i \in \{1, 2, \dots, r\}, \langle \mathbf{y}, \mathbf{a}_i \rangle \leq b_i\}$. Based on P ,

define the polyhedral cone

$$P' = \left\{ \begin{bmatrix} \mathbf{y} \\ y_{m+1} \end{bmatrix} \in \mathbb{R}^{m+1} : \left[\begin{array}{ccc|c} \text{---} & \mathbf{a}_1^\top & \text{---} & -b_1 \\ \text{---} & \mathbf{a}_2^\top & \text{---} & -b_2 \\ & \vdots & & \vdots \\ \text{---} & \mathbf{a}_r^\top & \text{---} & -b_r \\ \text{---} & \mathbf{0} & \text{---} & -1 \end{array} \right] \begin{bmatrix} \mathbf{y} \\ y_{m+1} \end{bmatrix} \leq \mathbf{0} \right\}.$$

Observe that $P \cong P'_1 := \{(\mathbf{y}, 1) \in \mathbb{R}^{m+1} : \mathbf{y} \in P\} \subseteq P'$. By Theorem 3.12, P' is finitely generated; let $V = \{\mathbf{u}'_1, \mathbf{u}'_2, \dots, \mathbf{u}'_s\}$ so that $P' = \text{cone}(V)$. For $i \in \{1, 2, \dots, s\}$, we write each $\mathbf{u}'_i = (\mathbf{u}_i, u_{i,m+1})$ where $\mathbf{u}_i \in \mathbb{R}^m$ and $u_{i,m+1} \in \mathbb{R}$. So we also have

$$P' = \left\{ \begin{bmatrix} \mathbf{y} \\ y_{m+1} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^s \lambda_i \mathbf{u}_i \\ \sum_{i=1}^s \lambda_i u_{i,m+1} \end{bmatrix} : \lambda_1, \lambda_2, \dots, \lambda_s \in \mathbb{R}_+ \right\}.$$

Since $y_{m+1} = \sum_{i=1}^s \lambda_i u_{i,m+1} \geq 0$ for any conic combination, we may conclude that $u_{i,m+1} \in \mathbb{R}_+$ for every $i \in \{1, 2, \dots, s\}$. We partition $\{1, 2, \dots, s\} = J_0 \cup J_+$ where $J_0 = \{j : u_{j,m+1} = 0\}$ and $J_+ = \{j : u_{j,m+1} > 0\}$; we define

$$\mu_i = \begin{cases} \lambda_i u_{i,m+1} & \text{if } i \in J_+ \\ \lambda_i & \text{if } i \in J_0 \end{cases} \quad \text{and} \quad \mathbf{v}_i = \begin{cases} \mathbf{u}_i / u_{i,m+1} & \text{if } i \in J_+ \\ \mathbf{u}_i & \text{if } i \in J_0 \end{cases}.$$

That is, if $u_{i,m+1} = 0$, then it annihilates the contribution, as well as the restriction on λ_i to y_{m+1} . We may verify

$$P'_1 = \left\{ \begin{bmatrix} \mathbf{y} \\ 1 \end{bmatrix} : \mathbf{y} = \sum_{i=1}^s \mu_i \mathbf{v}_i, \quad 1 = \sum_{i \in J_+} \mu_i, \quad \mu_1, \mu_2, \dots, \mu_s \in \mathbb{R}_+ \right\}. \quad (3.15) \{?\}$$

since every non-negative μ_i can be traced back to λ_i and, since we chose $v_{i,m+1} = 1$, $\lambda_i \in \mathbb{R}_+$ always. As the definition of P'_1 , we may write

$$P = \left\{ \sum_{j \in J_+} \mu_j \mathbf{v}_j : 1 = \sum_{i \in J_+} \mu_i, \quad \forall i \in \{1, 2, \dots, s\}, \mu_i \in \mathbb{R}_+ \right\} + \left\{ \sum_{j \in J_0} \mu_j \mathbf{v}_j : \forall i \in \{1, 2, \dots, s\}, \mu_i \in \mathbb{R}_+ \right\}. \quad (3.16) \text{eqn: MW constr}$$

For the converse direction, where we are given a finitely generated cone and a convex hull, observe that the above deduction is a sequence of equivalences. Thus, we proved both directions. $\square$

In a Minkowski-Weyl representation of a polyhedral set $P = \text{conv}(V) + C$ for some

set of vectors V , the cone C is known as the recession cone of P . Let X be a set in $\mathbb{R}^m$; a vector $\mathbf{d} \in \mathbb{R}$ is a *recession* of X if $\mathbf{x} + \alpha\mathbf{d} \in X$ for every $\mathbf{x} \in X$ and $\alpha \in \mathbb{R}_+$. By the universal quantifiers, the set of all recessions of X is well-defined; we denote the set of recessions of X by

$$\text{rec}(X) = \{\mathbf{d} \in \mathbb{R}^m : \forall \mathbf{x} \in X, \forall \alpha \in \mathbb{R}_+, \mathbf{x} + \alpha\mathbf{d} \in X\}.$$

In a way, $\text{cone}(X)$ is the minimal cone that contains X , and $\text{rec}(X)$ is a maximal translated cone whose translation is properly contained in X .

Corollary 3.17. *For a polyhedral set $P = \{\mathbf{x} \in \mathbb{R}^m : A\mathbf{x} \leq \mathbf{b}\}$ for $A \in \mathbb{R}^{n \times m}$ and $\mathbf{b} \in \mathbb{R}^n$ and its Minkowski-Weyl representation of P as in Equation (3.16) defined by $\mathbf{v}_i$ for $i \in \{1, 2, \dots, s\} = J_0 \cup J_+$ such that*

$$P = \text{conv}(V) + C := \text{conv}(\{\mathbf{v}_j : j \in J_+\}) + \text{cone}(\{\mathbf{v}_j : j \in J_0\});$$

we have that $C = \text{rec}(P)$.

Proof: Let $P, A, \mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r, \mathbf{b}, b_1, b_2, \dots, b_r, V, J_0$ be defined as in the statement; further, we take P' and P'_1 as in the proof of Theorem 3.14. We will first show that $C \subseteq \{\mathbf{d} : A\mathbf{d} \leq \mathbf{0}\}$. For $j \in J_0$, observe the construction that $\mathbf{v}_j$, the generators of C , are defined by $\mathbf{v}_j = \mathbf{u}_j$ where $(\mathbf{u}_j, 0) \in P'$; that is, $\langle \mathbf{a}_i, \mathbf{u}_j \rangle - u_{j,m+1}b_j \leq 0$ for every $i \in \{1, 2, \dots, r\}$. Since J_0 is chosen by $J_0 = \{j : u_{j,m+1} = 0\}$, we have $u_{j,m+1}b_j = 0$ for $j \in J_0$; equivalently, $A\mathbf{v}_j = A\mathbf{u}_j \leq \mathbf{0}$ for every generator of C . This concludes that every point in $\mathbf{d} \in C$ satisfies that $A\mathbf{d} \leq \mathbf{0}$.

Suppose that $\mathbf{d} \in C$ and we have that $A\mathbf{d} \leq \mathbf{0}$. For any point $\mathbf{x} \in P$ and any $\alpha \in \mathbb{R}_+$, we have that $A(\mathbf{x} + \alpha\mathbf{d}) = A\mathbf{x} + \alpha A\mathbf{d} \leq \mathbf{b} + \mathbf{0}$, which follows the definition of P , satisfies that $\mathbf{x} + \alpha\mathbf{d} \in P$ and $\mathbf{d} \in \text{rec}(P)$.

Suppose that $\mathbf{d} \in \text{rec}(P)$; by definition, for any point $\mathbf{x} \in P$ and any $\alpha \in \mathbb{R}_+$, we have that $A(\mathbf{x} + \alpha\mathbf{d}) = A\mathbf{x} + \alpha A\mathbf{d} \leq \mathbf{b}$. This holds only if $A\mathbf{d} \leq \mathbf{0}$ since, if any entry of $A\mathbf{d}$ is positive, then replacing α by a sufficiently large number α' violates the fact that $A(\mathbf{x} + \alpha'\mathbf{d}) \leq \mathbf{b}$. Indeed, if $(\mathbf{d}, 0) = \sum_{i=1}^s \lambda_i(\mathbf{v}_i, u_{i,m+1})$, then the last coordinate gives $0 = \sum_{j \in J_+} \lambda_j u_{j,m+1}$. For $i \in J_+$, since $u_{i,m+1} > 0$, we have $\lambda_i = 0$ and, thus, $\mathbf{d}$ can be written as a conic combination of $\{\mathbf{v}_j : j \in J_0\}$; as such $\mathbf{d} \in C$. $\square$

We make the last remark of the section. We make a simple observation that a polyhedral set on $\mathbb{R}^2$ is, eventually, drawn by drawing lines on a plane and considering the region that they surround. An intersection of two lines creates an angle. As a spoiler, these intersection that creates angles play a pivotal role in linear programming and the simplex method in Chapter 4. Let C be a convex set; an *extreme point* $\mathbf{b} \in K$ is a point where there is no line ℓ properly contained in K contains $\mathbf{b}$ in it; that is, for all points $\mathbf{c}, \mathbf{c}' \in K$ so that $\mathbf{b} \notin \{\mathbf{c}, \mathbf{c}'\}$, we have that $\mathbf{b} \neq \alpha\mathbf{c} + (1 - \alpha)\mathbf{c}'$ for all $\alpha \in (0, 1)$.

Theorem 3.18. *For a polyhedral set $P \subseteq \mathbb{R}^n$ and its Minkowski-Weyl Representation $P = \text{conv}(V) + C$ for some $V = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$; we have that every extreme point of P belongs to V .*

Proof: We proceed with a proof by contraposition; suppose that $\mathbf{x} \in P \setminus V$ and we will show that it is not an extreme point. Let $\mathbf{v} = \sum_{i=1}^m \mu_i \mathbf{v}_i \in \text{cone}(V)$ be a conic combination, $\mathbf{d} \in C$, and we write that $\mathbf{x} = \mathbf{v} + \mathbf{d}$. By Corollary 3.17, $C = \text{rec}(P)$. First, we may assume that $\mathbf{d} \neq \mathbf{0}$. By the definition of a recession, we have that $\mathbf{v} + 2\mathbf{d} \in P$ as well. This gives that

$$\mathbf{x} = \frac{1}{2}\mathbf{v} + \frac{1}{2}(\mathbf{v} + 2\mathbf{d}) \in P,$$

which states that $\mathbf{x}$ belongs to the line in P with endpoints $\mathbf{v}$ and $\mathbf{v} + 2\mathbf{d}$; thus, $\mathbf{x}$ is not an extreme point.

We now assume that $\mathbf{d} = \mathbf{0}$. We also have that $|V| > 1$ as, otherwise, $\mathbf{v} \in \text{conv}(\{\mathbf{v}_1\}) = \{\mathbf{v}_1\}$. As $|V| > 1$, without loss of generality, we assume that $0 \notin \{\mu_1, \mu_2\}$. By the definition of a convex set, let $\mu = \mu_1 + \mu_2$, for all $\alpha \in (0, \mu)$, $\alpha \mathbf{v}_1 + (\mu - \alpha) \mathbf{v}_2 + \sum_{i=3}^m \mu_i \mathbf{v}_i \in P$; we have that $\mathbf{x}$ is not an extreme point. $\square$

Consider the following example that seems to be a counterexample but actually is not.

Example 3.19. Consider the following two sets

$$V = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \quad \text{and} \quad R = \left\{ \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$$

that defines a polyhedral set $P = \text{conv}(V) + \text{cone}(R) \subseteq \mathbb{R}^2$. This polyhedral set can be simplified as the closed halfspace

$$\left\{ \mathbf{x} \in \mathbb{R}^2 : \left\langle \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{x} \right\rangle \leq 1 \right\},$$

which does not contain an extreme point even V consists of all extreme points of $\text{conv}(V)$.

The following proposition has a page-long, linear-algebraic proof; however, if we restrict to the case of polyhedral sets, then it is a natural corollary of the Minkowski-Weyl Theorem.

Proposition 3.20 (Krein-Milman Theorem). *If $K \subseteq \mathbb{R}^n$ is a closed convex set that is compact, then K is equal to the convex hull of its extreme points.*

Corollary 3.21. *If $P \subseteq \mathbb{R}^n$ is a compact polyhedral set, then P is equal to the convex hull of its extreme points.*

Proof: Let $P = \text{conv}(X) + C$ be a polyhedral set with its Minkowski-Weyl representation. A polyhedral set is closed and convex; if P is compact, then P is bounded, which means $C = \{\mathbf{0}\}$. Toward a contradiction, suppose $\mathbf{x} \in X$ is not an extreme point of P ; we then have some $\alpha \in (0, 1)$ and $\mathbf{u}, \mathbf{w} \in P$ such that $\mathbf{x} = \alpha\mathbf{u} + (1 - \alpha)\mathbf{w}$ and, with respect of X , the convex combination representation of $\mathbf{u}$ and $\mathbf{w}$ can be $\mathbf{x}$ -free (i.e., the coefficient of $\mathbf{x}$ in the representation is 0). Thus, $\text{conv}(X) = \text{conv}(X \setminus \{\mathbf{x}\})$. The statement follows by inductively removing all the non-extreme points. $\square$

3.2 Algebraic Characterization of Extreme Points

This subsection develops the fundamental linear-algebraic characterization of extreme points. Rather than evaluating extreme points solely through geometric convex combinations, the three theorems in this section provide direct, verifiable linear independence criteria tailored to general, standard, and bounded polyhedral representations. While established here algebraically, these characterizations serve as foundational anchors; subsequent sections will explicitly link each criterion back to its geometric, matrix-algebraic, and algorithmic interpretations. We first write down the theorem statements for the section and prove them in the following subsections.

me points part

1) **Theorem 3.22** (Extreme points and row spaces). *Suppose that $P \subseteq \mathbb{R}^m$ is a polyhedral set defined by $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \in \mathbb{R}^m$ and $b_1, b_2, \dots, b_n \in \mathbb{R}$, so that $P = \{\mathbf{x} : \forall i \in \{1, 2, \dots, n\}, \langle \mathbf{a}_i, \mathbf{x} \rangle \leq b_i\}$. A point $\mathbf{v} \in P$ is an extreme point of P if and only if the set*

$$S_{\mathbf{v}} = \{\mathbf{a}_j : j \in \{1, 2, \dots, n\}, \langle \mathbf{a}_j, \mathbf{v} \rangle = b_j\}$$

contains one or more subsets consisting of m linearly independent vectors.

Geometrically, Theorem 3.22 characterizes an extreme point as the unique solution that attains m linearly independent constraint hyperplanes in $\mathbb{R}^m$. In the following, we focus on a specific, yet fundamental class of polyhedral sets: those that are defined in the first orthant and those that are bounded. We will later see that these types of polyhedral sets, with their algebraic properties, have computational advantages. In Let $\mathbf{v} \in \mathbb{R}^n$ and v_i be the value on its i -th coordinate; the *support* of $\mathbf{v}$ is the index set $\text{supp}(\mathbf{v}) = \{i : v_i \neq 0\}$.

me points part

2) **Theorem 3.23** (Extreme points and column spaces). *Suppose $P \subseteq \mathbb{R}_+^m$ is a polyhedral set obtained by the intersection of finitely many hyperplanes, $P = \{\mathbf{x} : M\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$, where $M \in \mathbb{R}^{n \times m}$ and $\mathbf{b} \in \mathbb{R}^n$. A vector $\mathbf{v} \in P$ is an extreme point of P if and*

only if the columns of M with indices in $\text{supp}(\mathbf{v})$ are linearly independent.

Let $\mathbf{u}, \mathbf{w}, \mathbf{v} \in \mathbb{R}^m$, such that $\mathbf{u} \leq \mathbf{w}$, and u_i, w_i, v_i be their i -th entry, respectively; $\text{supp}_{\mathbf{u}, \mathbf{w}}(\mathbf{v}) = \{i : u_i < v_i < w_i\}$.

me points part

3) **Theorem 3.24** (Extreme points and column spaces in a bounded region). *Suppose we have a polyhedral set $P = \{\mathbf{x} \in \mathbb{R}^m : M\mathbf{x} = \mathbf{b}, \mathbf{c} \leq \mathbf{x} \leq \mathbf{d}\}$, where $M \in \mathbb{R}^{n \times m}$, $\mathbf{b} \in \mathbb{R}^n$, and $\mathbf{c}, \mathbf{d} \in \mathbb{R}^m$. A vector $\mathbf{v} \in P$ is an extreme point of P if and only if the columns of M with indices so that the corresponding entries in $\mathbf{v}$ that are strictly between $\mathbf{c}$ and $\mathbf{d}$ are linearly independent.*

Critically, these column space properties do not hold for arbitrary polyhedral sets without such one- or two-sided bounds.

3.2.1 Ranks of Active Constraints

Proof of Theorem 3.22: For the forward implication, let $\mathbf{v} \in \mathbb{R}^m$ be an extreme point of P and we denote $S = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$. Without loss of generality, suppose $S_{\mathbf{v}} = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_s\}$, for some $s \geq m$. Let A and $A_{\mathbf{v}}$ be the matrices obtained by stacking all the transposes of every element of S and $S_{\mathbf{v}}$, respectively. Toward a contradiction, suppose that $S_{\mathbf{v}}$ contains fewer than m linearly independent vectors. By the contradictory hypothesis, the kernel of $A_{\mathbf{v}}$ is nontrivial, i.e., $\ker(A_{\mathbf{v}}) \neq \{\mathbf{0}\}$; let $\bar{\mathbf{w}} \in \ker(A_{\mathbf{v}})$ be a nonzero point. Since A is a linear function, we have that $A(\mathbf{v} + \gamma\bar{\mathbf{w}}) = A\mathbf{v} + \gamma A\bar{\mathbf{w}}$ for any $\gamma \in \mathbb{R}$. By the definition of $S_{\mathbf{v}}$ and by the choice of $\bar{\mathbf{w}}$, we have that $\langle \mathbf{a}_j, \mathbf{v} + \gamma\bar{\mathbf{w}} \rangle = b_j + \gamma \cdot 0 = b_j$ if $\mathbf{a}_j \in S_{\mathbf{v}}$ and $\langle \mathbf{a}_i, \mathbf{v} \rangle < b_i$ if $\mathbf{a}_i \in S \setminus S_{\mathbf{v}}$; this means there is some $\epsilon_i \in \mathbb{R}_{++}$ for all $i \in \{i : \mathbf{a}_i \in S \setminus S_{\mathbf{v}}\}$ so that $\langle \mathbf{a}_i, \mathbf{v} \rangle + \epsilon_i = b_i$. Let $\epsilon = \max_{i \in \{1, 2, \dots, s\}} \epsilon_i$; we may choose $\gamma \in \mathbb{R}_{++}$ so that, for all $\mathbf{a}_i \in S \setminus S_{\mathbf{v}}$, we have that $\langle \mathbf{a}_i, \mathbf{v} \rangle \pm \gamma \langle \mathbf{a}_i, \bar{\mathbf{w}} \rangle \leq b_i$ and, thus, $A(\mathbf{v} \pm \gamma\bar{\mathbf{w}}) \leq \mathbf{b}$. Yet, this yields that the two points $\mathbf{v} + \gamma\bar{\mathbf{w}}, \mathbf{v} - \gamma\bar{\mathbf{w}} \in P$, which contradicts that $\mathbf{v}$ is an extreme point of P .

For the converse direction, suppose $\mathbf{v} \in P$ defines $S_{\mathbf{v}}$ and $S' \subseteq S_{\mathbf{v}}$ is a subset consisting of m linearly independent vectors. Suppose $\mathbf{u}, \mathbf{w} \in P$ are points such that $\alpha \in (0, 1)$ so that $\mathbf{v} = \alpha\mathbf{u} + (1 - \alpha)\mathbf{w}$; note that $\langle \mathbf{u}, \mathbf{a}_j \rangle \leq b_j$ and $\langle \mathbf{w}, \mathbf{a}_j \rangle \leq b_j$. Now, for any $\mathbf{a}_j \in S'$ we have that

$$\begin{aligned} b_j = \langle \mathbf{v}, \mathbf{a}_j \rangle &= \langle \alpha\mathbf{u} + (1 - \alpha)\mathbf{w}, \mathbf{a}_j \rangle \\ &= \alpha \langle \mathbf{u}, \mathbf{a}_j \rangle + (1 - \alpha) \langle \mathbf{w}, \mathbf{a}_j \rangle \\ &\leq \alpha b_j + (1 - \alpha) b_j \\ &= b_j. \end{aligned}$$

By a *strict weighted-average* argument, we have that $\langle \mathbf{u}, \mathbf{a}_j \rangle = \langle \mathbf{w}, \mathbf{a}_j \rangle = b_j$.

Let A' be the matrix obtained by stacking the transposes of vectors $\mathbf{a}_j \in S'$, and let $\mathbf{b}'$ be the vector so that the b_j are stacked in the same order. Since S' consists of n linearly

independent solutions, we have that A' is a full-rank matrix and, if there is a solution such that $A'\mathbf{x} = \mathbf{b}'$, then the solution is unique; thus, we have $\mathbf{u} = \mathbf{v} = \mathbf{w}$. Thus, we have that $\mathbf{v}$ is an extreme point. $\square$

With respect to a point $\mathbf{x}^*$, a constraint $\mathbf{a}$ is said to be *active* if $\langle \mathbf{a}, \mathbf{x}^* \rangle = b_i$.

Definition 3.25. Let P be a polyhedral set in $\mathbb{R}^m$ defined by a list of constraints and bounds $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_r, \mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \in \mathbb{R}^m$ and $q_1, q_2, \dots, q_r, b_1, b_2, \dots, b_n \in \mathbb{R}$ so that

$$P = \{\mathbf{x} : \forall i \in \{1, 2, \dots, r\}, \langle \mathbf{e}_i, \mathbf{x} \rangle = q_i, \\ \forall j \in \{1, 2, \dots, n\}, \langle \mathbf{a}_j, \mathbf{x} \rangle \leq b_j \}.$$

Let $\mathbf{x}^* \in \mathbb{R}^m$.

- The point $\mathbf{x}^*$ is a *basic solution* if all the equality constraints are active and there are m linearly independent active constraints with respect to $\mathbf{x}^*$.
- The point $\mathbf{x}^*$ is a *basic feasible solution* if $\mathbf{x}$ is a basic solution and satisfies all the constraints.

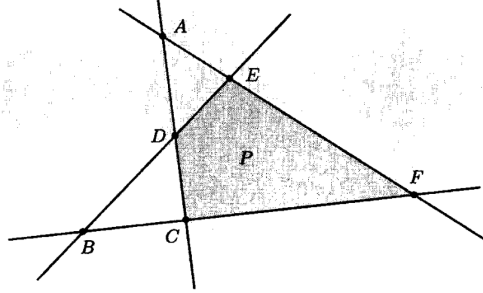


Figure 3.1: Figure 2.7 in *Introduction To Linear Optimization* by Bertsekas and Tsitsiklis. The figure illustrates a polyhedral set $P \subseteq \mathbb{R}^2$ where A, B, C, D, E, F are all basic solutions to P but only C, D, E, F are basic feasible solutions. Every pair on the same line is a pair of adjacent basic solutions to P .

We should remind junior readers that the term “basic” stands for the adjective for “basis”, which states that the solution is generated by a chosen linear algebraic basis; we will see more discussion in the following chapters. In the following, we consider a linear algebraic property of the intersection of a convex set and a hyperplane and the behavior of an extreme point under intersections.

3.2.2 Basic Feasible Solutions for Standard Form

We denote the unit vector with the i -th entry being 1 by $\chi_i \in \mathbb{R}^m$. Let $B \subseteq \{1, 2, \dots, m\}$ be an index set and $\mathbf{v} \in \mathbb{R}^m$; we write $[\mathbf{v}]_B$ to be the vector in $\mathbb{R}^{|B|}$ obtained from $\mathbf{v}$ by

taking the entries with indices in K .

Proof of Theorem 3.23: Suppose that $P \subseteq \mathbb{R}_+^m$ is obtained by the intersection of finitely many hyperplanes, $P = \{\mathbf{x} : M\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$ for some $M \in \mathbb{R}^{n \times m}$ and $\mathbf{b} \in \mathbb{R}^n$. We rewrite $\mathbf{x} \in P$ as a solution to the system of linear inequalities

$$A\mathbf{x} = \begin{bmatrix} M \\ -M \\ -I_m \end{bmatrix} \mathbf{x} \leq \begin{bmatrix} \mathbf{b} \\ -\mathbf{b} \\ \mathbf{0} \end{bmatrix}.$$

For the forward direction, suppose $\mathbf{v} \in \mathbb{R}_+^m$ is an extreme point in P and, toward contradiction, the columns in M corresponding to the indices in $\text{supp}(\mathbf{v})$ are not linearly independent. By the definition of linear independence, there then is a vector $\mathbf{d} \in \mathbb{R}^m$ where the entries satisfy that

$$\text{supp}(\mathbf{d}) \subseteq \text{supp}(\mathbf{v}), \quad M\mathbf{d} = \mathbf{0}, \quad \text{and} \quad \mathbf{d} \neq \mathbf{0},$$

which gives that, for any $\epsilon \in \mathbb{R}_{++}$,

$$A(\mathbf{v} + \epsilon\mathbf{d}) = \begin{bmatrix} M \\ -M \\ -I_m \end{bmatrix} (\mathbf{v} + \epsilon\mathbf{d}) = \begin{bmatrix} M\mathbf{v} \\ -M\mathbf{v} \\ -I_m\mathbf{v} \end{bmatrix} + \epsilon \begin{bmatrix} M\mathbf{d} \\ -M\mathbf{d} \\ -I_m\mathbf{d} \end{bmatrix} = \begin{bmatrix} M\mathbf{v} \\ -M\mathbf{v} \\ -\mathbf{v} \end{bmatrix} + \epsilon \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ -\mathbf{d} \end{bmatrix}$$

Since $\text{supp}(\mathbf{d}) \subseteq \text{supp}(\mathbf{v})$ for a non-zero $\mathbf{d}$, we have $\mathbf{v} \neq \mathbf{0}$ as well. We now choose

$$\epsilon = \min_{i \in \text{supp}(\mathbf{d})} \frac{v_i}{|d_i|}$$

where v_i and d_i are the i -th component of $\mathbf{v}$ and $\mathbf{d}$, respectively, so that $-\mathbf{v} \pm \epsilon\mathbf{d} \leq \mathbf{0}$. From the first two blocks, we have that $M(\mathbf{v} \pm \epsilon\mathbf{d}) = M\mathbf{v}$ and, thus, $\mathbf{v} \pm \epsilon\mathbf{d} \in P$, which contradicts that $\mathbf{v}$ is an extreme point.

For the converse direction, suppose $\mathbf{v} \in P$ and the columns corresponding to $\text{supp}(\mathbf{v})$ are linearly independent. Let $k = |\text{supp}(\mathbf{v})|$ and, without loss of generality, $\text{supp}(\mathbf{v}) = \{1, 2, \dots, k\}$; we denote by $M_k \in \mathbb{R}^{n \times k}$ consists of the first k columns of M . Since M_k then has precisely k linearly independent columns and, thus, has a rank of k or more, by the fundamental theorem of linear algebra, there is a list of k rows of M_k that are linearly independent after transposing; we denote the set of indices of these rows by J . We then take the rows of M with indices in J and denote the transpose of these rows by $S_J = \{\mathbf{a}_j \in \mathbb{R}^m : j \in J\}$.

Let $R_v = \{\chi_j \in \mathbb{R}^m : j \in \{1, 2, \dots, m\} \setminus \text{supp}(\mathbf{v})\}$, which corresponds to rows in $-I_m$. Suppose that there are $\mu_j, \lambda_i \in \mathbb{R}$ for $j \in J$ and $i \in \{1, 2, \dots, m\} \setminus \text{supp}(\mathbf{v})$ so that

$$\mathbf{0} = \sum_{j \in J} \mu_j \mathbf{a}_j + \sum_{i \in \{1, 2, \dots, m\} \setminus \text{supp}(\mathbf{v})} \lambda_i \chi_i; \tag{3.26} \quad \boxed{\text{eqn: column sp}}$$

we will now show that $S_J \cup R_{\mathbf{v}}$ is linearly independent by showing that $\{\mu_j : j \in J\} \cup \{\lambda_i : i \in \{1, 2, \dots, m\} \setminus \text{supp}(\mathbf{v})\} = \{0\}$. We first consider the computation within the entries with indices in $\text{supp}(\mathbf{v})$. Since we chose S_J to be the “rows” that are linearly independent in M_k , the entries with indices $\text{supp}(\mathbf{v})$ form vectors in $\mathbb{R}^k$ that are linearly independent; in particular, we have

$$\begin{aligned} & \left[\sum_{j \in J} \mu_j \mathbf{a}_j + \sum_{i \in \{1, 2, \dots, m\} \setminus \text{supp}(\mathbf{v})} \lambda_i \boldsymbol{\chi}_i \right]_{\text{supp}(\mathbf{v})} \\ &= \sum_{j \in J} \mu_j [\mathbf{a}_j]_{\text{supp}(\mathbf{v})} + \sum_{i \in \{1, 2, \dots, m\} \setminus \text{supp}(\mathbf{v})} \lambda_i [\boldsymbol{\chi}_i]_{\text{supp}(\mathbf{v})} \\ &= \sum_{j \in J} \mu_j [\mathbf{a}_j]_{\text{supp}(\mathbf{v})} \end{aligned}$$

and $[\mathbf{0}]_{\text{supp}(\mathbf{v})} = \sum_{j \in J} \mu_j [\mathbf{a}_j]_{\text{supp}(\mathbf{v})}$ implies that $\mu_j = 0$ for all $j \in J$. Thus, this annihilates a summation in Equation (3.26) so that $\mathbf{0} = \sum_{i \in \{1, 2, \dots, m\} \setminus \text{supp}(\mathbf{v})} \lambda_i \boldsymbol{\chi}_i$, which is obvious that $\lambda_i = 0$ for all $i \in \{1, 2, \dots, m\} \setminus \text{supp}(\mathbf{v})$. This concludes that $S_J \cup R_{\mathbf{v}}$ is a set of linearly independent vectors.

Finally, we have $\langle \mathbf{a}_j, \mathbf{v} \rangle \leq b_i$ for $j \in J$ and, since $\mathbf{v} \in P$, we have $-\langle \mathbf{a}_j, \mathbf{v} \rangle \leq -b_i$, which implies that $\langle \mathbf{a}_j, \mathbf{v} \rangle = b_i$. For $R_{\mathbf{v}}$ consists of $\boldsymbol{\chi}_i$ with $i \in \{1, 2, \dots, m\} \setminus \text{supp}(\mathbf{v})$ (i.e., the coordinates with a 0-entry in $\mathbf{v}$), we have $\langle \mathbf{v}, \boldsymbol{\chi}_i \rangle = 0$. Thus, along with $S_J \cup R_{\mathbf{v}}$ consisting of m linearly independent rows in M , we apply Theorem 3.22 so that $\mathbf{v}$ is an extreme point. $\square$

Finally, we may prove Theorem 3.24, which is essentially the bounded version of Theorem 3.23. Though the proofs are parallel, the proof of Theorem 3.24 employs an argument identical to the proof for Theorem 3.23 with some minor twists in the setup.

Proof of Theorem 3.24: The polyhedral set defines the following system of linear inequality

$$\begin{bmatrix} M \\ -M \\ -I_m \\ I_m \end{bmatrix} \mathbf{x} \leq \begin{bmatrix} \mathbf{b} \\ -\mathbf{b} \\ -\mathbf{u} \\ \mathbf{w} \end{bmatrix}$$

For the forward direction, suppose that $\mathbf{v}$ is an extreme point and, toward a contradiction, suppose that the columns of M corresponding to $\text{supp}_{\mathbf{u}, \mathbf{w}}(\mathbf{v})$ are linearly dependent, which gives a vector $\mathbf{d} \in \mathbb{R}^m$ so that

$$M\mathbf{d} = \mathbf{0}, \quad \mathbf{d} \neq \mathbf{0}, \quad \text{and} \quad \text{supp}_{\mathbf{u}, \mathbf{w}}(\mathbf{d}) \subseteq \text{supp}_{\mathbf{u}, \mathbf{w}}(\mathbf{v})$$

For any $\epsilon \in \mathbb{R}_{++}$, we have

$$A(\mathbf{v} + \epsilon \mathbf{d}) = \begin{bmatrix} M\mathbf{v} \\ -M\mathbf{v} \\ -\mathbf{v} \\ \mathbf{v} \end{bmatrix} + \epsilon \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ -\mathbf{d} \\ \mathbf{d} \end{bmatrix}$$

and choose small enough

$$\epsilon = \min_{i \in \text{supp}_{\mathbf{u}, \mathbf{w}}(\mathbf{v})} \frac{\min(\{v_i - u_i, w_i - v_i\})}{|d_i|}$$

and it would contradict that $\mathbf{v}$ is an extreme point.

For the converse direction, replacing supp by $\text{supp}_{\mathbf{u}, \mathbf{w}}$ and one may verify that all the linear independence conditions remain valid. $\square$

3.3 Existence of Extreme Points

We examine and emphasize the existence of extreme points in a finitely generated polyhedral set. Recall the hypothesis in Theorem 3.23 and 3.24 that we are given a polyhedral set with only equalities. We say a polyhedral set $P \subseteq \mathbb{R}^n$ is in *standard form* if P is defined by equality constraints $M\mathbf{x} = \mathbf{b}$ and inequality constraints $P \subseteq \mathbb{R}_+^n$; that is, if $M \in \mathbb{R}^{m \times n}$, then P is defined with $m + n$ constraints. In a later chapter, we will see why a program in standard form is necessary and sufficient for constructing a solution. Observe that, if P is a polyhedral set in standard form, then a point is a basic solution if and only if it is a basic feasible solution.

Lemma 3.27. *Let $K \subseteq \mathbb{R}^n$ be a convex set and $H \subseteq \mathbb{R}^n$ be a hyperplane such that $K \cap H \neq \emptyset$. If $K \not\subseteq H$, then $\dim(K \cap H) \leq \dim(K) - 1$.*

Proof: Observe that $\text{aff}(K \cap H) \subseteq \text{aff}(K)$, we have that $\dim(K \cap H) \leq \dim(K)$. Since H itself is also an affine set, we have $\text{aff}(K \cap H) \subseteq H$. Toward a contradiction, suppose that $\dim(K \cap H) = \dim(K)$. Yet, since $\text{aff}(K \cap H)$ and $\text{aff}(K)$ are uniquely defined by subspaces, we have that $\text{aff}(K \cap H) = \text{aff}(K)$, which then gives that $\text{aff}(K) \subseteq H$. Now, $K \subseteq \text{aff}(K)$, we have $K \subseteq H$, which contradicts the assumption that $K \not\subseteq H$. $\square$

Theorem 3.28. *Let $K \subseteq \mathbb{R}^n$ be a nonempty convex set. Let H be a hyperplane so that one of its closed halfspaces contains K . Every extreme point in $K \cap H$ is an extreme point in K .*

Proof: Toward a contradiction, suppose a point $\mathbf{x}^* \in K \cap H$ is an extreme point but is not an extreme point in K . There then are distinct points $\mathbf{u}, \mathbf{w} \in K \setminus \{\mathbf{x}^*\}$ and

$\alpha \in (0, 1)$ so that $\mathbf{x}^* = \alpha \mathbf{u} + (1 - \alpha) \mathbf{w}$. Since $\mathbf{x}^* \in H$, there is a vector $\mathbf{a}$ normal to H that defines $H = \{\mathbf{y} : \langle \mathbf{y}, \mathbf{a} \rangle = \langle \mathbf{x}^*, \mathbf{a} \rangle\}$; further, we assume that for all $\mathbf{x} \in K$, we have $\langle \mathbf{x}, \mathbf{a} \rangle \geq \langle \mathbf{x}^*, \mathbf{a} \rangle$ that K is contained in this halfspace of H . This gives that $\langle \mathbf{u}, \mathbf{a} \rangle \geq \langle \mathbf{x}^*, \mathbf{a} \rangle$ and $\langle \mathbf{w}, \mathbf{a} \rangle \geq \langle \mathbf{x}^*, \mathbf{a} \rangle$, which in view of $\mathbf{x}^* = \alpha \mathbf{u} + (1 - \alpha) \mathbf{w}$ implies that equality holds for both. Therefore, we have that $\mathbf{u}, \mathbf{w} \in K \cap H$, but this contradicts the assumption that $\mathbf{x}^*$ is an extreme point in $K \cap H$. $\square$

Recall that, given a convex set X , a vector $\mathbf{d}$ is a recession of X if, for any $\mathbf{x} \in X$ and $\alpha \in \mathbb{R}_+$, we have that $\mathbf{x} + \alpha \mathbf{d} \in X$, and $\text{rec}(X)$ is the collection of all recessions of X . We first review some more properties of $\text{rec}(X)$ and then finish this section by proving a critical characterization, Theorem 3.31.

ion cone theorem

Proposition 3.29 (Recession Cone Theorems). *Let X be a closed, convex, and nonempty set in $\mathbb{R}^n$.*

1. $\text{rec}(X)$ is a closed and convex cone.
2. A vector $\mathbf{d}$ belongs to $\text{rec}(X)$ if and only if there is a point $\mathbf{x} \in X$ such that $\mathbf{x} + \alpha \mathbf{d} \in X$ for all $\alpha \in \mathbb{R}_+$.
3. If there is a nonzero direction in $\text{rec}(X)$, then X is unbounded.
4. $\text{rec}(\text{ri}(X)) = \text{rec}(X)$.
5. Let Y be a closed and convex set so that $X \cap Y \neq \emptyset$, we have that $\text{rec}(X \cap Y) = \text{rec}(X) \cap \text{rec}(Y)$; this is closed under arbitrary intersection.

Proof: We only prove Item (2), and we treat the remaining items as given since they involve too much point-set topology. For the forward direction for Item (2), it follows from the definition that a direction $\mathbf{d} \in \text{rec}(X)$ such that $\mathbf{x} + \alpha \mathbf{d} \in X$ for all $\mathbf{x} \in X$ and $\alpha \in \mathbb{R}_+$.

For the converse direction, suppose that $\mathbf{d} \in \mathbb{R}^n$ and there is some $\mathbf{x}_0 \in X$ so that $\mathbf{x}_0 + \alpha_0 \mathbf{d} \in X$ for all $\alpha_0 \in \mathbb{R}_+$. Fix a point $\mathbf{y} \in X \setminus \{\mathbf{x}_0\}$ and a scalar $\alpha \in \mathbb{R}_+$ arbitrarily; we will show that $\mathbf{y} + \alpha \mathbf{d} \in X$. Starting from $\mathbf{x}_0 + \alpha \mathbf{d}$, which is an element of X , consider the line

$$\left\{ \mathbf{z}_\lambda := (1 - \lambda)(\mathbf{y} + \alpha \mathbf{d}) + \lambda(\mathbf{x}_0 + \alpha \mathbf{d}) \quad : \quad \lambda \in [0, 1] \right\}.$$

Observe the rearrangement for $\lambda \in (0, 1]$ so that

$$\begin{aligned} \mathbf{z}_\lambda &= (1 - \lambda)(\mathbf{y} + \alpha \mathbf{d}) + \lambda(\mathbf{x}_0 + \alpha \mathbf{d}) \\ &= \mathbf{y} + \alpha \mathbf{d} - \lambda \mathbf{y} - \lambda \alpha \mathbf{d} + \lambda \mathbf{x}_0 + \lambda \alpha \mathbf{d} \\ &= \mathbf{y} + \alpha \mathbf{d} - \lambda \mathbf{y} + \lambda \mathbf{x}_0 \\ &= (1 - \lambda) \mathbf{y} + \lambda \left(\mathbf{x}_0 + \frac{\alpha}{\lambda} \mathbf{d} \right); \end{aligned}$$

in particular, we may interpret $\mathbf{z}_\lambda$ as belonging to the line segment connecting $\mathbf{y}$ and $\mathbf{x}_0 + (\alpha/\lambda)\mathbf{d}$. As $\alpha/\lambda \in \mathbb{R}_+$, we have that $\mathbf{y}, \mathbf{x}_0 + (\alpha/\lambda)\mathbf{d} \in X$ so $\mathbf{z}_\lambda \in X$ by the convexity of X . By the convexity of X , That says, for all $\lambda \in (0, 1]$, we have $\mathbf{z}_\lambda \in X$.

Finally, defined with the index set $I = (\lambda_k)_{k \in \mathbb{N}_+}$ where $\lambda_k = 1/k$, we have a sequence

$$\left\{ \mathbf{z}_{\frac{1}{k}} := \mathbf{y} + \alpha \mathbf{d} + \frac{1}{k}(\mathbf{x}_0 - \mathbf{y}) : \lambda_k \in I \right\} \subseteq X;$$

this is easy to verify that it is a Cauchy sequence and approaches $\mathbf{y} + \alpha \mathbf{d}$. Since X is closed, every Cauchy sequence is convergent and, thus, we have that $\mathbf{y} + \alpha \mathbf{d} \in X$. $\square$

Previously we discussed a “strictly hyperplane separation theorem”, Proposition 1.7; this is too strong for the next theorem, so we study the idea of separation again and provide a weaker condition. Recall that a hyperplane in $\mathbb{R}^n$ is defined with a vector $\mathbf{a}$ and a scalar b so that $H = H_{\mathbf{a},b} = \{\mathbf{x} \in \mathbb{R}^n : \langle \mathbf{a}, \mathbf{x} \rangle = b\}$. Two sets X and Y are said to be *separated* by H if $\langle \mathbf{x}, \mathbf{a} \rangle \leq b \leq \langle \mathbf{y}, \mathbf{a} \rangle$ for all $\mathbf{x} \in X$ and $\mathbf{y} \in Y$ or $\langle \mathbf{y}, \mathbf{a} \rangle \leq b \leq \langle \mathbf{x}, \mathbf{a} \rangle$ for all $\mathbf{x} \in X$ and $\mathbf{y} \in Y$; without loss of generality, we assume the prior case. We may further write that $\sup_{\mathbf{x} \in X} \langle \mathbf{x}, \mathbf{a} \rangle \leq \inf_{\mathbf{y} \in Y} \langle \mathbf{y}, \mathbf{a} \rangle$.

We say a hyperplane H *properly separates* two sets X and Y if H separates X and Y and $X \cup Y \not\subseteq H$. For a singleton $\{\mathbf{x}\}$ we may H separates $\mathbf{x}$ and Y if H separates $\{\mathbf{x}\}$ and Y . For example, consider the following “intervals on the x -axis”

$$I_{0,1} = \left\{ \begin{bmatrix} x \\ 0 \end{bmatrix} : x \in [0, 1] \right\} \quad \text{and} \quad I_{2,3} = \left\{ \begin{bmatrix} y \\ 0 \end{bmatrix} : y \in [2, 3] \right\}$$

that, in fact, can be separated by the “ x -axis”: Consider the unit vector with a 1 on the second coordinate, $\chi_2 \in \mathbb{R}^2$; we have $X = \{\mathbf{x} : \langle \chi_2, \mathbf{x} \rangle = 0\}$ and, indeed for every $\mathbf{x} \in I_{0,1}$ and $\mathbf{y} \in I_{2,3}$, we have $\langle \chi_2, \mathbf{x} \rangle = \langle \chi_2, \mathbf{y} \rangle = 0$ and, thus, H separates $I_{0,1}$ and $I_{2,3}$. However, this separation is geometrically meaningless. Thus, we need the notion of proper separation to pick a line with a slope to separate $I_{0,1}$ and $I_{2,3}$.

proper separation

Proposition 3.30 (Proper Separation Theorem). *Let $C \subseteq \mathbb{R}^n$ be a nonempty convex set and $\mathbf{x} \in \mathbb{R}^n$. There exists a hyperplane that properly separates C and $\mathbf{x}$ if and only if $\mathbf{x} \notin \text{ri}(C)$.*

reme iff no line

Theorem 3.31. *Suppose X is a closed and non-empty convex set. The set X has one or more extreme point if and only if X does not contain a line.*

Proof: Suppose $\mathbf{x}^* \in X$ is an extreme point and there is also some line $L = \{\mathbf{a} + \lambda \mathbf{d} : \lambda \in \mathbb{R}\}$ where $\mathbf{a} \in X$ and $\mathbf{d} \neq \mathbf{0}$; namely, both $\mathbf{d}, -\mathbf{d} \in \text{rec}(X)$. By the recession cone theorem, Theorem 3.29 Item 2, we have that $\mathbf{x}^* + \lambda \mathbf{d} \in X$ for all $\lambda \in \mathbb{R}$; but this contradicts the fact that $\mathbf{x}^*$ does not belong to a line as an extreme point.

For the converse direction, we assume that X does not contain a line. We proceed with a proof by induction on the dimension of the affine set of X . Suppose $\dim(\text{aff}(X)) = 0$, which means X is a singleton, and the only point is the extreme point automatically.

Inductively, suppose that for any X_0 with $\dim(\text{aff}(X_0)) < k$ in which X_0 does not contain a line and contains one or more extreme points. Suppose X has that $\dim(\text{aff}(X)) = k$. Since an affine set is a translated subspace, which always contains a line, we have that $X \neq \text{aff}(X)$. Thus, we may take a point from $\tilde{x} \in X \setminus \text{ri}(X)$. By the proper hyperplane separation theorem, Theorem 3.30, there is a hyperplane H in $\text{aff}(X)$ that contains $\text{ri}(X)$ in one of its halfspaces and properly separates $\tilde{x}$ and $\text{ri}(X)$. We take $X \cap H$, since $\text{ri}(X) \neq \emptyset$, we know $X \not\subseteq H$ and, thus, $\dim(\text{aff}(X \cap H)) < \dim(\text{aff}(X))$ by Lemma 3.27. As X and H are both closed, we have that $X \cap H$ is closed, non-empty, and contains no line. We have that $X \cap H$ contains one or more extreme points. By Theorem 3.28, an extreme point in $X \cap H$ remains an extreme point in X , which proves the theorem that X has one or more extreme points. $\square$

Chapter 4

Linear Programming

ar programming) In this section, we consider the celebrated algorithm for computing an optimal solution in a linear program. The bedrock of this chapter is the collection of Theorem 3.22, 3.23, and 3.24. In particular, this method is defined only for polyhedra defined on a one- or two-sided bounded polyhedron.

4.1 Fundamental Theorem of Linear Programming

We take the following theorem as given. Let $f : X \rightarrow \mathbb{R}$ and $X^* = \{x^* \in X : f(x^*) = \inf_{x \in X} f(x)\}$. Observe that, if f is not bounded from below, then $X^* = \emptyset$; for its contraposition, if $X^* \neq \emptyset$, then f is bounded from below.

implies constant) **Proposition 4.1.** *Let f be a convex function over X and $X \subseteq \mathbb{R}^n$ such that $X \neq \emptyset$. Let $X^* \subseteq X$ whose elements are $x \in X^*$ attain the minimum over X . If $X^* \cap \text{ri}(X) \neq \emptyset$, then f is constant over X ; that is, $X = X^*$.*

?(Bauer) **Theorem 4.2** (Bauer's Extreme Point Theorem for Convex Maximization). *Let f be a convex function defined on a closed convex set $C \subseteq \mathbb{R}^n$, where C contains one or more extreme points. Let $C^* \subseteq C$ be the set of elements that attain the maximum of f . If $C^* \neq \emptyset$, then C^* contains one or more extreme points of C .*

Proof: We proceed with induction on the dimension of C . Suppose that $\dim(C) = 0$, then C contains only a single point, which is an extreme point. Meanwhile, the only point in C naturally attains the maximum of f .

Inductively, suppose that $\dim(C) = r > 0$ and that every convex function f_0 defined on a convex set $C_0 \subseteq \mathbb{R}^n$, with dimension $r - 1$ or less and contains one or more extreme points, has that, if $C_0^* \neq \emptyset$, then C_0^* contains one or more extreme points of C_0 , where C_0^* is the set of points in C_0 that attain the maximum of f_0 over C_0 .

Let $\mathbf{x}^* \in C^*$. By Proposition 4.1, if $\mathbf{x}^* \in \text{ri}(C)$, then f is constant over C and, thus, $C^* = C$ and we are done since C contains one or more extreme point. Otherwise, we have that $\mathbf{x}^* \in C \setminus \text{ri}(C)$. By Theorem 3.30 and Proposition 1.11 (i.e., the proper separation theorem and that convex sets have nonempty relative interior), there is a hyperplane H that separates $\mathbf{x}^*$ from $\text{ri}(C)$; since $\mathbf{x}^* \in C$, we also have $\mathbf{x}^* \in C \cap H$. By Lemma 3.27, we have that $\dim(C \cap H) < \dim(C)$ and $f|_{C \cap H}$ remains convex. Apply Theorem 3.31 and $C \cap H \subseteq C$ and C does not contain a line subspace and, thus, neither does $C \cap H$ contain a line; therefore, $C \cap H$ contains one or more extreme points by Theorem 3.31 again.

Inductively, let C_1^* be the set of elements that attain the maximum of $f|_{C \cap H}$; since $\mathbf{x}^* \in C^*$, we have $\mathbf{x}^* \in C_1^*$, which implies that elements in C_1^* attain the minimum of f over C , and $C_1^* \subseteq C^*$. By induction, as $\dim(C \cap H) < \dim(C)$, we have that C_1^* contains one or more extreme points of $C \cap H$. By Theorem 3.28, which states that the extreme point in $C \cap H$ is an extreme point in C , we have that C^* contains one or more extreme points in C , which was what we wanted. $\square$

Theorem 4.3 (Fundamental Theorem of Linear Programming). *Given $\mathbf{c} \in \mathbb{R}^n$ and a polyhedral set $P \subseteq \mathbb{R}^n$ with one or more extreme points, consider a linear function $\langle \mathbf{c}, \cdot \rangle$ defined over P . Either f is not bounded from above, or there is an extreme point in P that attains the maximum $\sup_{\mathbf{x} \in P} f(\mathbf{x})$.* $\square$

4.2 Linear Algebraic Characterization of Polyhedra

Given a polyhedral set P , a point $\mathbf{x} \in \mathbb{R}^n$ is called a *vertex in P* if $\mathbf{x} \in P$ and there is some $\mathbf{c} \in \mathbb{R}^n$ so that $\langle \mathbf{c}, \mathbf{x} \rangle > \langle \mathbf{c}, \mathbf{x}' \rangle$ for all $\mathbf{x}' \in P \setminus \{\mathbf{x}\}$. Observe that the definition of a vertex and the definition of an extreme point and a basic feasible solution are the same object in three different aspects. Before unifying the three characterizations of a polyhedral corner, it is worth reflecting on the logical architecture of our exposition. In Chapter 3, we established the fundamental equivalence between extreme points and basic feasible solutions (extreme point $\iff$ basic feasible solution). This ordering is both deliberate and necessary: extreme points (geometric/topological) and basic feasible solutions (algebraic/algorithmic) are intrinsic to the polyhedron P itself, depending solely on its defining constraint system without reference to an objective function. By isolating this intrinsic geometry-to-algebra bridge first in Theorem 3.22, we established a constructive computational skeleton for P . In this section, we introduce the extrinsic optimization perspective: the notion of a vertex defined relative to an optimization objective $\langle \mathbf{c}, \cdot \rangle$; and close the loop to synthesize all three viewpoints into one unified equivalence.

thm: unification

Theorem 4.4 (The Unification Theorem). *For a polyhedral set $P \subseteq \mathbb{R}^n$ and a point $\mathbf{x}^* \in \mathbb{R}_+^n$. The following are equivalent.*

1. The point $\mathbf{x}^*$ is a vertex in P .
2. The point $\mathbf{x}^*$ is an extreme point in P .
3. The point $\mathbf{x}^*$ is a basic feasible solution to P .

Proof: Theorem 3.22 shows that the point is an extreme point if and only if it is a basic feasible solution to P . We first show that a point $\mathbf{x}^*$ is a vertex in P implies that it is an extreme point in P ; next, we show that a point is a basic feasible solution to P implies that it is a vertex in P .

Suppose that $\mathbf{x}^* \in P$ is a vertex. By definition, there exists $\mathbf{c} \in \mathbb{R}^n$ such that for all $\mathbf{x} \in P \setminus \{\mathbf{x}^*\}$, we have that $\langle \mathbf{c}, \mathbf{x}^* \rangle > \langle \mathbf{c}, \mathbf{x} \rangle$. This says, for any two other points $\mathbf{u}, \mathbf{w} \in P$ and any $\alpha \in (0, 1)$, we have that

$$\langle \mathbf{c}, \mathbf{x}^* \rangle > \alpha \langle \mathbf{c}, \mathbf{u} \rangle + (1 - \alpha) \langle \mathbf{c}, \mathbf{w} \rangle.$$

Thus, there are no lines with endpoints in P that contain $\mathbf{x}^*$.

Suppose that $\mathbf{x}^* \in P$ is a basic feasible solution to P . Without loss of generality, we may assume $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_r$ are linear independent active constraints so that $\langle \mathbf{a}_j, \mathbf{x}^* \rangle = b_j$ that attains the bounds b_j , for $j \in \{1, 2, \dots, r\}$. We define a vector $\mathbf{c} = \sum_{i=1}^m \mathbf{a}_i$. Since the constraints are active, we have that $\langle \mathbf{c}, \mathbf{x}^* \rangle = \sum_{i=1}^m b_i$. By the linear independence, $\mathbf{x}^*$ is the unique solution so that $\langle \mathbf{a}_j, \mathbf{x}^* \rangle = b_j$ for all $j \in \{1, 2, \dots, r\}$ and, for any other $\mathbf{x} \in P \setminus \{\mathbf{x}^*\}$, we have that $\langle \mathbf{a}_j, \mathbf{x}^* \rangle > \langle \mathbf{a}_j, \mathbf{x} \rangle$; this follows the definition of $\mathbf{x}^*$ is a vertex in P . $\square$

From Theorem 4.4, an extreme point defined geometrically now has an intrinsic and linear-algebraic characterization as a basic feasible solution, and an extrinsic definition as a vertex. Theorem 3.31 provided a geometric characterization of the existence of an extreme point; the next theorem provides a linear-algebraic characterization of the existence of an extreme point.

Theorem 4.5. *Let $P \subseteq \mathbb{R}^n$ be a polyhedral set defined with m constraints and bounds $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m \in \mathbb{R}^n$ and $b_1, b_2, \dots, b_m \in \mathbb{R}$.*

1. *The polyhedral set P has one or more extreme point;*
2. *The polyhedral set P contains no line;*
3. *There is a list of n linearly independent constraints.*

Proof: As Theorem 3.31 showed $(1) \iff (2)$, it suffices to show that $(1) \implies (3)$ and $(3) \implies (2)$. For $(1) \implies (3)$, suppose that there is an extreme point $\mathbf{x}^* \in P$; by Theorem 4.4, it is also a basic feasible solution in P and, by definition, there are n linearly independent active constraints with respect to $\mathbf{x}^*$.

For $(3) \implies (2)$, without loss of generality, suppose $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ is a list of n linearly independent constraints. Toward a contradiction, suppose there is a line $\ell = \{\mathbf{x} + \alpha \mathbf{d} : \alpha \in \mathbb{R}\} \subseteq P$ for some $\mathbf{x}, \mathbf{d} \in \mathbb{R}^n$ and $\mathbf{d} \neq \mathbf{0}$. By the definition of the polyhedral set P , we

have

$$\langle \mathbf{a}_i, \mathbf{x} + \alpha \mathbf{d} \rangle = \langle \mathbf{a}_i, \mathbf{x} \rangle + \alpha \langle \mathbf{a}_i, \mathbf{d} \rangle \leq b_i \quad \text{for } i \in \{1, 2, \dots, n\} \text{ and } \alpha \in \mathbb{R},$$

which implies that $\langle \mathbf{a}_i, \mathbf{d} \rangle = 0$ since, otherwise, the inequalities is violated by choosing α positively large enough or negatively large enough. Since $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \in \mathbb{R}^n$ is a list of linearly independent vector and $\langle \mathbf{a}_i, \mathbf{d} \rangle = 0$ for all $i \in \{1, 2, \dots, n\}$, we have that $\mathbf{d} = 0$, which contradicts the assumption that ℓ is a line. $\square$

4.3 Optimal Conditions and Degeneracy

We now consider the “basic” part of a basic feasible solution. Recall that a point $\mathbf{x} \in \mathbb{R}^n$ is a basic solution to $P \subseteq \mathbb{R}^n$ if all the equality constraints are active and there are n active constraints at $\mathbf{x}$ in total. In this section, we provide more insights on viewing a linear program with basic- and basic feasible solutions. We first restate Theorem 3.23, which shows the equivalence between an extreme point and a basic feasible solution; we alter the notation for the indices to fit the convention in this chapter.

Theorem (3.23 restated). *Suppose $P \subseteq \mathbb{R}_+^n$ is a polyhedral set obtained by the intersection of finitely many hyperplanes, $P = \{\mathbf{x} : M\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$, where $M \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. A vector $\mathbf{v} \in P$ is an extreme point of P if and only if the columns of M with indices in $\text{supp}(\mathbf{v})$ are linearly independent.*

In the following, we would assume the linear program lives in $\mathbb{R}^n$ and is defined by a matrix $M \in \mathbb{R}^{m \times n}$ with $m \leq n$; in particular, the assumption $m \leq n$ is given by the following fact.

Theorem 4.6. *Let P be a polyhedral set defined by equality constraints $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m \in \mathbb{R}^n$ and bounds $b_1, b_2, \dots, b_m \in \mathbb{R}$ so that $P = \{\mathbf{x} \in \mathbb{R}^n : \forall i \in \{1, 2, \dots, m\} \langle \mathbf{a}_i, \mathbf{x} \rangle = b_i\}$. Without loss of generality, suppose that $\dim(\text{span}(\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m\})) = k < m$ and $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ is a list of linearly independent vectors. Define $Q = \{\mathbf{x} \in \mathbb{R}^n : \forall i \in \{1, 2, \dots, k\}, \langle \mathbf{a}_i, \mathbf{x} \rangle = b_i\}$, we have that $Q = P$. $\square$*

For a matrix $M \in \mathbb{R}^{m \times n}$ and an index set $B \subseteq \{1, 2, \dots, n\}$ of order m , i.e. $|B| = m$; for convenience, we denote the i -th least index in B by $B(i)$, for $i \in \{1, 2, \dots, m\}$. With respect to B , if the list of $B(1)$ -th, $B(2)$ -th, $\dots$, $B(m)$ -th columns of M are linearly independent, then we call them as the *basic columns*. In terms of solving a system of linear equations $M\mathbf{x} = \mathbf{b}$, we call the set of $B(i)$ -th variables of $\mathbf{x}$ the *basic variables*; we call the columns in M and the variables not in B as *nonbasic*. Generally, we denote $M_B \in \mathbb{R}^{m \times m}$ as the submatrix of M whose i -th column being the $B(i)$ -th column of M , for $i \in \{1, 2, \dots, m\}$.

Suppose B corresponds to m linearly independent columns of $M \in \mathbb{R}^{m \times n}$ and, thus, M_B is a full-rank $m \times m$ -matrix. The sub-system of equations of $M\mathbf{x} = \mathbf{b}$ then has a unique solution $[\mathbf{x}]_B \in \mathbb{R}^m$ to

$$M_B[\mathbf{x}]_B = \mathbf{b} \quad \text{or, equivalently} \quad [\mathbf{x}]_B = M_B^{-1}\mathbf{b};$$

recall that $[\mathbf{x}]_B$ indicates the subvector extracted from entries of $\mathbf{x}$ with indices in B .

Consider a point $\mathbf{x}$ so that the entries with their indices in B are formed by $[\mathbf{x}]_B$ and 0 otherwise. Observe that, $\text{supp}(\mathbf{x}) \subseteq B$ since not all entries in $[\mathbf{x}]_B$ are non-zero. By Theorem 3.23 and 4.4, we have that $\mathbf{x} \in P$ is a basic solution. For a polyhedral set in standard form, being basic and being basic feasible are equivalent.

Remark 4.7. The above operation to obtain $\mathbf{x}$ is sometimes called the construction of a basic solution. Some textbooks refer to it as an “algorithm starting by **choosing** (or **picking**) m linearly independent columns.” However, the last author of this booklet felt that it is misleading since the verb “choosing” refers to an explicit algorithm that should be introduced beforehand. However, the “choosing” part is implicitly given in the later procedure, implied by theorems instead of being explicitly extracted from a matrix; thus, we assume the linearly independent vectors are given for now.

Assumption 4.8. Throughout the section, we consider the same linear program and notations defined as follows. Let $\mathbf{c} \in \mathbb{R}^n$, $M \in \mathbb{R}^{m \times n}$, and $\mathbf{b} \in \mathbb{R}^m$, define the linear program

$$\begin{aligned} & \max_{\mathbf{x} \in \mathbb{R}^n} \langle \mathbf{c}, \mathbf{x} \rangle \\ & \text{subject to} \quad \mathbf{x} \in P \\ & \text{where} \quad P = \{\mathbf{x} : M\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\} \end{aligned} \tag{LPS}$$

is the standard form polyhedral set defined by the system of equalities and inequalities. Let B be an index set of a list of m linearly independent columns in M and $N = \{1, 2, \dots, n\} \setminus B$. Denote M by its column form so that $M = [\mathbf{f}_1 \mid \mathbf{f}_2 \mid \dots \mid \mathbf{f}_n]$ and $M_B = [\mathbf{f}_{B(1)} \mid \mathbf{f}_{B(2)} \mid \dots \mid \mathbf{f}_{B(m)}]$ be the full-rank $m \times m$ -submatrix. We denote each entry of $\mathbf{c}$ as c_j for $j \in \{1, 2, \dots, n\}$.

In the following, we consider the “elbow room” of a basic solution within P . For convenience, let the j -th entry of a vector $\mathbf{x}$ be x_j ; we denote $[\mathbf{x}]_j = [\mathbf{x}]_{\{j\}}$.

Definition 4.9. Consider (LPS) in Assumption 4.8. For each nonbasic index $k \in N$, define the k -th basic direction $\mathbf{d}^{(k)} \in \mathbb{R}^n$ with respect to B , with each of the j -th entry

of $\mathbf{d}^{(k)}$, as the following: define

$$[\mathbf{d}^{(k)}]_B = -M_B^{-1} \mathbf{f}_k, \quad [\mathbf{d}^{(k)}]_{N \setminus \{k\}} = \mathbf{0}, \quad \text{and} \quad [\mathbf{d}^{(k)}]_k = 1.$$

Define j -th reduced cost as

$$\bar{c}_j = c_j - \langle [\mathbf{c}]_B, M_B^{-1} \mathbf{f}_j \rangle$$

and form the reduced cost $\bar{\mathbf{c}}$ by the collection of all the j -th reduced cost.

Using the notations that are consistent with those in Definition 4.9, we remark on a few points. As M_B has full rank, the matrix M_B is invertible; so $[\mathbf{d}^{(k)}]_B$ is well-defined. For $j \in B$ defined by $j = B(i)$ for some $i \in \{1, 2, \dots, m\}$, the indicator vector $\boldsymbol{\chi}_i$ has that $M_B \boldsymbol{\chi}_i = \mathbf{f}_{B(i)}$ and, equivalently, $\boldsymbol{\chi}_i = M_B^{-1} \mathbf{f}_{B(i)}$; this is a reason for that basis direction is not defined for indices in B . For $j \in B$, we then have $\bar{c}_j = c_j - \langle [\mathbf{c}]_B, \boldsymbol{\chi}_{B(i)} \rangle = c_j - c_j = 0$ since, again, that M_B is invertible and bijective. Otherwise, when $j \in N$, we have $M_B^{-1} \mathbf{f}_j = -[\mathbf{d}^{(j)}]_B$ is the j -th basic direction. That is, fix a basis index B , we have

$$\bar{c}_j = \begin{cases} c_j - \langle [\mathbf{c}]_B, M_B^{-1} \mathbf{f}_j \rangle & \text{if } j \in N, \\ 0 & \text{otherwise.} \end{cases}$$

Finally, consider the vectorization that $\bar{\mathbf{c}}$ can be written in the form of

$$\bar{\mathbf{c}} = \mathbf{c} - \left([\mathbf{c}]_B^\top M_B^{-1} M \right)^\top = \mathbf{c} - (M_B^{-1} M)^\top [\mathbf{c}]_B.$$

Lemma 4.10. Consider (LPS) in Assumption 4.8. For $k \in \{1, 2, \dots, m\} \setminus B$, let $\mathbf{d}^{(k)} \in \mathbb{R}^n$ be the k -th basic direction with respect to B . We have that $M \mathbf{d}^{(k)} = \mathbf{0}$.

Proof: We denote B by its column form so that $M = [\mathbf{f}_1 \mid \mathbf{f}_2 \mid \dots \mid \mathbf{f}_n]$ and let d_j denote the j -th entry of $\mathbf{d}^{(k)}$, for $j \in \{1, 2, \dots, n\}$. Note that a matrix multiplication $M \mathbf{d}^{(k)} = \mathbf{0}$ can be reinterpreted as a linear combination. For each $k \in N$, we obtain that

$$\begin{aligned} M \mathbf{d}^{(k)} &= \sum_{j=1}^n d_j \mathbf{f}_j \\ &= \sum_{j \in B} d_j \mathbf{f}_j + \sum_{j \in N} d_j \mathbf{f}_j \\ &= M_B [\mathbf{d}^{(k)}]_B + d_k \mathbf{f}_k + \sum_{j \in N \setminus \{k\}} d_j \mathbf{f}_j. \end{aligned}$$

By the definition that B chooses a column basis of M , the matrix M_B is invertible and,

thus, $[\mathbf{d}^{(k)}]_B = -M_B^{-1}\mathbf{f}_k$ and $d_k = 1$ and $d_j = 0$ otherwise, we continue so that

$$\begin{aligned}
M\mathbf{d}^{(k)} &= M_B[\mathbf{d}^{(k)}]_B + d_k\mathbf{f}_k + \sum_{j \in N \setminus \{k\}} d_j\mathbf{f}_j \\
&= M_B(-M_B^{-1}\mathbf{f}_k) + 1 \cdot \mathbf{f}_k + \sum_{j \in N \setminus \{k\}} 0 \cdot \mathbf{f}_j \\
&= -\mathbf{f}_k + \mathbf{f}_k \\
&= \mathbf{0}.
\end{aligned}$$

This finishes the proof. □

Theorem 4.11. Consider (LPS) in Assumption 4.8 and let $\mathbf{x}^* \in \mathbb{R}^n$ be a basic feasible solution in P . Suppose that B also satisfies that $\text{supp}(\mathbf{x}^*) \subseteq B$ and $[\mathbf{x}^*]_N = \mathbf{0}$. If the reduced cost $\bar{\mathbf{c}}$ associated with B satisfies that $\bar{\mathbf{c}} \leq \mathbf{0}$, then $\mathbf{x}^*$ attains the maximum.

Proof: Let $\mathbf{x} = \mathbf{x}^*$. Suppose $\mathbf{y} \in P \setminus \{\mathbf{x}\}$ is another basic solution in P ; we have that $M\mathbf{x} = M\mathbf{y} = \mathbf{b}$ and $\mathbf{y} - \mathbf{x} \neq \mathbf{0}$; consider the directional vector $\mathbf{u} = \mathbf{y} - \mathbf{x}$ that has $M\mathbf{u} = \mathbf{0}$. Denote $M = [\mathbf{f}_1 \mid \mathbf{f}_2 \mid \cdots \mid \mathbf{f}_n]$ and u_j, x_j, y_j denotes the j -th entries of $\mathbf{u}, \mathbf{x}, \mathbf{y}$, for $j \in \{1, 2, \dots, n\}$, to show that $\mathbf{x}$ is optimal, we show that

$$\langle \mathbf{c}, \mathbf{y} \rangle - \langle \mathbf{c}, \mathbf{x} \rangle = \langle \mathbf{c}, \mathbf{u} \rangle = \sum_{j=1}^n c_j u_j = \langle [\mathbf{c}]_B, [\mathbf{u}]_B \rangle + \sum_{j \in N} c_j u_j \leq 0$$

Consider the rearrangement $\mathbf{0} = M\mathbf{u} = M_B[\mathbf{u}]_B + \sum_{j \in N} u_j \mathbf{f}_j$, which gives that

$$[\mathbf{u}]_B = -\sum_{j \in N} u_j M_B^{-1} \mathbf{f}_j.$$

We then have that

$$\begin{aligned}
\langle \mathbf{c}, \mathbf{u} \rangle &= \langle [\mathbf{c}]_B, [\mathbf{u}]_B \rangle + \sum_{j \in N} c_j u_j \\
&= \left\langle [\mathbf{c}]_B, -\sum_{j \in N} u_j M_B^{-1} \mathbf{f}_j \right\rangle + \sum_{j \in N} c_j u_j \\
&= \sum_{j \in N} c_j u_j - \sum_{j \in N} u_j \langle [\mathbf{c}]_B, M_B^{-1} \mathbf{f}_j \rangle \\
&= \sum_{j \in N} (c_j - \langle [\mathbf{c}]_B, M_B^{-1} \mathbf{f}_j \rangle) u_j \\
&= \sum_{j \in N} \bar{c}_j u_j.
\end{aligned}$$

By the hypothesis that $[\mathbf{x}]_N = \mathbf{0}$ and $P \subseteq \mathbb{R}_+^n$, for $j \in N$, we have that $u_j = y_j - x_j = y_j - 0 = y_j \geq 0$. Since $\bar{\mathbf{c}} \leq 0$, we have that $\langle \bar{\mathbf{c}}, \mathbf{u} \rangle = \sum_{j \in N} \bar{c}_j u_j \leq 0$. $\square$

Let $P \subseteq \mathbb{R}_+^n$ be a polyhedral set defined by a matrix $M \in \mathbb{R}^{m \times n}$. Recall that a point $\mathbf{x}$ is a basic solution if there are n constraints that are active at $\mathbf{x}$. As the assumption is that $m \leq n$, if n constraints are active, then there are $n - m$ or more entries in $\mathbf{x}$ that attain the non-negative constraint, i.e., are zero. In other words, $\text{supp}(\mathbf{x}) \leq m$. A basic solution $\mathbf{x}$ is *degenerate* if there are more than n constraints active at $\mathbf{x}$.

degeneracy equiv

Remark 4.12. Equivalently, a basic solution $\mathbf{x}$ is degenerate if $|\text{supp}(\mathbf{x})| < m$. In particular, a basic solution $\mathbf{x}$ is non-degenerate if $|\text{supp}(\mathbf{x})| = m$ and, by Theorem 3.23, $\text{supp}(\mathbf{x})$ corresponds to a list of m linearly independent columns in M .

well-definedness

Lemma 4.13. *The reduced cost of a non-degenerate basic solution in a polyhedral set is well-defined.*

Proof: Consider (LPS) in Assumption 4.8 with a basic feasible solution $\mathbf{x}^*$. By Theorem 3.23, the indices in $\text{supp}(\mathbf{x}^*)$ correspond to linearly independent columns of M . Moreover, by the definition of degeneracy and Remark 4.12, the number of active constraints coincides with the cardinality of $|\text{supp}(\mathbf{x}^*)|$. Thus, under the definition of reduced cost, the full-rank submatrix induced from the standard form polyhedral set from the active constraints is unique, which implies the well-definedness. $\square$

direction shift

Lemma 4.14. *Consider (LPS) in Assumption 4.8 and let $\mathbf{x}$ be a non-degenerate basic feasible solution so that $B = \text{supp}(\mathbf{x}^*)$. For every $k \in \{1, 2, \dots, n\} \setminus B$, there exists $\theta \in \mathbb{R}_{++}$ so that so that $\mathbf{x} + \theta \mathbf{d}^{(k)} \in P$, where $\mathbf{d}^{(k)}$ is the k -th basic direction.*

Proof: By Lemma 4.13, the set of basic indices $B = \text{supp}(\mathbf{x})$ and the invertible submatrix M_B are uniquely defined. Note that $\mathbf{d}^{(k)}$ is defined by $[\mathbf{d}^{(k)}]_B = M_B^{-1} \mathbf{f}_k$, $d_k = 1$, and $d_j = 0$ for $j \in N \setminus \{k\}$, where $\mathbf{f}_k$ is the k -th column of M and d_k and d_j are the entries of $\mathbf{d}^{(k)}$ with indices in N . Since we have $\mathbf{x}$ is non-degenerate, $[\mathbf{x}]_B > \mathbf{0}$ and $[\mathbf{x}]_N = \mathbf{0}$. We now have $\mathbf{x} + \theta \mathbf{d}^{(k)}$ is formed by

$$\begin{aligned} [\mathbf{x} + \theta \mathbf{d}^{(k)}]_B &= [\mathbf{x}]_B - \theta M_B^{-1} \mathbf{f}_k, \\ [\mathbf{x} + \theta \mathbf{d}^{(k)}]_k &= x_k + \theta d_k = x_k + \theta, \text{ and} \\ [\mathbf{x} + \theta \mathbf{d}^{(k)}]_{N \setminus \{k\}} &= [\mathbf{x}]_{N \setminus \{k\}}. \end{aligned}$$

By Lemma 4.10, we have $M(\mathbf{x} + \theta \mathbf{d}^{(k)}) = M\mathbf{x}$; thus, as long as we maintain $\mathbf{x} + \theta \mathbf{d}^{(k)} \geq \mathbf{0}$, we have that $\mathbf{x} + \theta \mathbf{d}^{(k)} \in P$. Since we have restricted $\theta > 0$, we need to only control the entries with indices in B . If $\mathbf{d}^{(k)} \geq \mathbf{0}$, then we have $\mathbf{x} + \theta \mathbf{d}^{(k)} \in P$ for all $\theta \in \mathbb{R}_{++}$.

Otherwise, we may assume there is one or more indices in B so that $[\mathbf{d}^{(k)}]_i < 0$. Let x_j and d_j denote the j -th entry of $\mathbf{x}$ and $\mathbf{d}^{(k)}$, respectively, for $j \in B$. Define

$$\theta^* = \min_{i \in B, d_i < 0} \frac{x_i}{-d_i};$$

let $i^* \in B$ be the one that attains the minimum. By the choice, we have that

$$x_j + \left(-\frac{x_{i^*}}{d_{i^*}}\right) d_j \geq x_j + \left(-\frac{x_j}{d_j}\right) d_j \geq 0,$$

so $\mathbf{x} + \theta \mathbf{d}^{(k)} \in P$, which proves the statement. $\square$

mal reduced cost

Theorem 4.15. *Consider (LPS) in Assumption 4.8. If $\mathbf{x}^*$ is a nondegenerate basic feasible solution and attains the maximum, then the reduced cost $\bar{\mathbf{c}}$ has that $\bar{\mathbf{c}} \leq \mathbf{0}$.*

Proof: Suppose $\mathbf{x}$ is a nondegenerate basic solution and is an optimal solution for (LPS). By Lemma 4.13, there is a unique index set B of cardinality m so that M_B is an invertible submatrix. Toward a contradiction, suppose that $\bar{\mathbf{c}} \not\leq \mathbf{0}$; that is, there is some $k \in \{1, 2, \dots, n\}$ so that $c_k > 0$.

Recall that $c_j = 0$ for all $j \in B$; we have that $k \in N$. By Lemma 4.14, let $\theta \in \mathbb{R}_{++}$ be an appropriate scalar so that $\mathbf{x}^* + \theta \mathbf{d}^{(k)} \in P$. Consider

$$\langle \mathbf{c}, \mathbf{x}^* + \theta \mathbf{d}^{(k)} \rangle = \langle \mathbf{c}, \mathbf{x}^* \rangle + \theta \langle \mathbf{c}, \mathbf{d}^{(k)} \rangle.$$

We now have

$$\langle \mathbf{c}, \mathbf{d}^{(k)} \rangle = \langle [\mathbf{c}]_B, [\mathbf{d}^{(k)}]_B \rangle + c_k = c_k - \langle [\mathbf{c}]_B, M_B^{-1} \mathbf{f}_k \rangle = \bar{c}_k$$

By the contradictory hypothesis that $\bar{c}_k > 0$, along with $\theta > 0$, we have

$$\langle \mathbf{c}, \mathbf{x}^* + \theta \mathbf{d}^{(k)} \rangle = \langle \mathbf{c}, \mathbf{x}^* \rangle + \theta \bar{c}_k > \langle \mathbf{c}, \mathbf{x}^* \rangle,$$

which contradicts the hypothesis that $\mathbf{x}^*$ attains the maximum. $\square$

Under (LPS) in Assumption 4.8, by definition, we have $\mathbf{x} \in P$ implies that $M_B[\mathbf{x}]_B = \mathbf{b}$ and $[\mathbf{x}]_B \geq \mathbf{0}$. Now, since M_B is invertible, we have that $[\mathbf{x}]_B = M_B^{-1} \mathbf{b}$; the two parts in the necessary condition can be combined into $M_B^{-1} \mathbf{b} \geq \mathbf{0}$. Meanwhile, observe that the specification in the sufficient condition of Theorem 4.11 is to specify a basis since the assumption on degeneracy is missing; thus, if we include it, we may solely discuss the basis submatrix of the constraint matrix, independent of a specific point in P . As such, consider the following definition.

Definition 4.16. Consider (LPS) in Assumption 4.8. The matrix M_B is *optimal* if it satisfies that $M_B^{-1}\mathbf{b} \geq \mathbf{0}$ and the reduced cost $\bar{\mathbf{c}} = \mathbf{c} - (M_B^{-1}M)^\top[\mathbf{c}]_B \leq \mathbf{0}$.

Now, we are equipped with all the ground to implement Phase II of Dantzig's simplex method.

4.4 Simplex Method: the Algorithmic Procedure

As an algorithm, the simplex method takes in a linear program. The algorithm also optionally takes in a basic feasible solution to initiate the procedure or, if none is given, proceeds with a preliminary procedure to generate one; we will discuss this procedure in the next section. Thus, we study the so-called *Phase II* of the simplex method in this section; in particular, Phase II also provides intuition for which basic feasible solution to use if it is not provided.

4.4.1 The Non-Degenerate Conditions

Based on (LPS) in Assumption 4.8, we first assume there is a basic feasible solution $\mathbf{x}$ that is nondegenerate and define the unique B and induce the invertible matrix M_B . Compute each k -th reduced cost with respect to basis B and form $\mathbf{c}$; by Theorem 4.11, if $\bar{\mathbf{c}} \leq \mathbf{0}$, then $\mathbf{x}$ attains the maximum of $\langle \mathbf{c}, \cdot \rangle$ over P .

We now suppose that there is some k so that $\bar{c}_k > 0$, which repeats exactly the proof of Theorem 4.15, and we know that there is some $\theta \in \mathbb{R}_{++}$ so that $\mathbf{x} + \theta \mathbf{d}^{(k)} \in P$; in particular, Theorem 4.15 relies on Lemma 4.14 for the choice of θ . We now show that this choice leads to a basic feasible solution with respect to another basis.

Theorem 4.17. Consider (LPS) in Assumption 4.8 and let $\mathbf{x}$ be a nondegenerate basic feasible solution so that $B = \text{supp}(\mathbf{x})$. Suppose there is $k \in N$ such that $\bar{c}_k > 0$; either the maximization problem (LPS) is not bounded from above over P , or there is some $i^* \in B$ so that another point

$$\mathbf{x}' = \mathbf{x} - \frac{[\mathbf{x}]_{i^*}}{[\mathbf{d}^{(k)}]_{i^*}} \mathbf{d}^{(k)} \in P$$

is a basic feasible solution and $\langle \mathbf{c}, \mathbf{x}' \rangle > \langle \mathbf{c}, \mathbf{x} \rangle$.

Proof: By Theorem 4.15, $\mathbf{x}$ is not optimal. By the choice of $\mathbf{x}'$ in Lemma 4.14 and the constructive proof, if $\mathbf{d}^{(k)} \geq \mathbf{0}$, then the optimization problem $\max_{\mathbf{x} \in P} \langle \mathbf{c}, \mathbf{x} \rangle$ is not bounded from above.

We now assume that there is some $i \in N$ so that $[\mathbf{d}^{(k)}]_i < 0$; thus, we may choose $\theta = \min_{i \in B, d_i < 0} -x_i/d_i$ and denote $i^* \in B$ be the one that attains the minimum. Now,

we have that

$$[\mathbf{x}]_{i^*} - \frac{[\mathbf{x}]_{i^*}}{[\mathbf{d}^{(k)}]_{i^*}} [\mathbf{d}^{(k)}]_{i^*} = 0.$$

Let $\mathbf{x}' = \mathbf{x} + \theta \mathbf{d}^{(k)}$; we now need to further investigate the entries of $\mathbf{x}$ and $\mathbf{x}'$; without loss of generality, we may assume $i^* = 1$, $B = \{1, 2, \dots, m\}$, $k = m + 1$, and $N = \{m + 1, \dots, n\}$ so we write

$$\mathbf{x}' = \begin{bmatrix} [\mathbf{x}]_B \\ 0 \\ \mathbf{0} \end{bmatrix} + \theta \begin{bmatrix} [\mathbf{d}^{(k)}]_B \\ 1 \\ \mathbf{0} \end{bmatrix}$$

It is now clear that we have designed to have $\text{supp}(\mathbf{x}') \setminus \{k\} \subseteq \text{supp}(\mathbf{x}) \setminus \{i^*\}$ since the choice of i^* may not be unique but $[\mathbf{x}']_{i^*} = 0$ necessarily. Let $B' = B \setminus \{i^*\} \cup \{k\}$; we will show that $B' = \{2, 3, \dots, m, m + 1\}$ corresponds to a list of linearly independent columns of M . Let $(\mu_j)_{j \in B'}$ be a list of real numbers so that $\mathbf{0} = \sum_{j \in B'} \mu_j \mathbf{f}_j$; we apply M_B^{-1} to both sides of elements in $\mathbb{R}^m$ so that

$$\begin{aligned} \mathbf{0} = M_B^{-1} \mathbf{0} &= M_B^{-1} (\mu_2 \mathbf{f}_2 + \mu_3 \mathbf{f}_3 + \dots + \mu_{m+1} \mathbf{f}_{m+1}) \\ &= \mu_2 M_B^{-1} \mathbf{f}_2 + \mu_3 M_B^{-1} \mathbf{f}_3 + \dots + \mu_{m+1} M_B^{-1} \mathbf{f}_{m+1} \\ &= \mu_2 \boldsymbol{\chi}_2 + \mu_3 \boldsymbol{\chi}_3 + \dots + \mu_m \boldsymbol{\chi}_m + \mu_{m+1} [\mathbf{d}^{(k)}]_B \end{aligned} \tag{4.18}$$

where the first $m - 1$ terms are due to the observation that $M_B^{-1} \mathbf{f}_j = \boldsymbol{\chi}_j$ for $j \in B$ and the last term is given by the choice of k gives that $M_B^{-1} \mathbf{f}_k = -[\mathbf{d}^{(k)}]_B$. Since we have assumed $[\mathbf{d}^{(k)}]_{i^*} < 0$, where $i^* = 1$ and, in (4.18), all $\boldsymbol{\chi}_i$ for $i \in \{2, 3, \dots, m\}$ does not contribute to the first entry, we may conclude that $\mu_{m+1} = 0$. Now, since the list $\boldsymbol{\chi}_2, \boldsymbol{\chi}_3, \dots, \boldsymbol{\chi}_m$ is linearly independent, we have that $\{\mu_2, \mu_3, \dots, \mu_m\} = \{0\}$. This concludes that $\boldsymbol{\chi}_2, \boldsymbol{\chi}_3, \dots, \boldsymbol{\chi}_m, [\mathbf{d}^{(k)}]_B$ is a list of linearly independent vectors and, by properties of invertible mappings, $\mathbf{f}_2, \mathbf{f}_3, \dots, \mathbf{f}_{m+1}$ is a list of linearly independent vectors. By Theorem 3.23 and Theorem 4.4, $\mathbf{x}'$ is a basic feasible solution. $\square$

Step 1

For each $k \in B$,

4.5 Initiating the Simplex Method

4.5.1 Reduction to Standard Form

We first show that every polyhedral set can be reduced into a standard form polyhedral set. Without loss of generality, every polyhedral set has the following form: let

$\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m \in \mathbb{R}^n$ and $b_1, b_2, \dots, b_m \in \mathbb{R}$; define

$$P = \{\mathbf{x} \in \mathbb{R}^n : \forall j \in \{1, 2, \dots, m\}, \langle \mathbf{a}_j, \mathbf{x} \rangle \leq b_j\}. \quad (4.19) \quad \boxed{\text{poly simplex}}$$

For each constraint $\mathbf{a}_j$, for $j \in \{1, 2, \dots, m\}$, introduce a *slack variable* $s_j \in \mathbb{R}_+$ and form a vector $\mathbf{s} \in \mathbb{R}_+^m$; consider

$$P_1 = \left\{ \begin{bmatrix} \mathbf{x} \\ \mathbf{s} \end{bmatrix} \in \mathbb{R}^{n+m} : \forall i \in \{1, 2, \dots, m\}, \langle \mathbf{a}_j, \mathbf{x} \rangle + s_j = b_j, s_j \geq 0 \right\}.$$

Observe that every feasible solution $\mathbf{x}_0 \in P$, there is some $\mathbf{s}_0 \in \mathbb{R}^m$ so that $(\mathbf{x}_0, \mathbf{s}_0) \in P_1$; conversely, for every $(\mathbf{x}_0, \mathbf{s}_0) \in P_1$, we have $\mathbf{x}_0 \in P$.

Observe that a number can always be represented by the difference of two non-negative numbers; for example, $5 = 5 - 0$ and $-5 = 0 - 5$. We start by assuming that none of the variables of $\mathbf{x}$ has bounds on them. For each variable x_i , where $i \in \{1, 2, \dots, n\}$, introduce two variable x_i^+ and x_i^- and form $\mathbf{x}^+, \mathbf{x}^- \in \mathbb{R}^n$. Consider a third polyhedral set such that

$$P_2 = \left\{ \begin{bmatrix} \mathbf{x}^+ \\ \mathbf{x}^- \\ \mathbf{s} \end{bmatrix} \in \mathbb{R}^{2n+m} : \begin{array}{ll} \forall j \in \{1, 2, \dots, n\}, & \langle \mathbf{a}_j, \mathbf{x}^+ \rangle - \langle \mathbf{a}_j, \mathbf{x}^- \rangle + s_j = b_j, \\ & s_j \geq 0, \\ \mathbf{x}^+ \geq \mathbf{0}, \\ \mathbf{x}^- \geq \mathbf{0}. \end{array} \right\} \quad (4.20) \quad \boxed{\text{poly simplex s}}$$

Observe that if $(\mathbf{x}_0^+, \mathbf{x}_0^-, \mathbf{s}_0) \in P_2$, then $(\mathbf{x}_0^+ - \mathbf{x}_0^-, \mathbf{s}_0) \in P_1$. Conversely, suppose $(\mathbf{x}_0, \mathbf{s}_0) \in P_1$; let x_i denote the i -th entry of $\mathbf{x}_0$, where $i \in \{1, 2, \dots, n\}$, set pairs

$$(x_j^+, x_j^-) = \begin{cases} (x_j, 0) & \text{if } x_j \geq 0. \\ (0, -x_j) & \text{otherwise,} \end{cases}$$

and define $\mathbf{x}_0^+$ and $\mathbf{x}_0^-$ accordingly, we have $\mathbf{x}_0^+, \mathbf{x}_0^- \in \mathbb{R}_+^n$ and $(\mathbf{x}_0^+, \mathbf{x}_0^-, \mathbf{s}_0) \in P_2$. Without loss of generality, we may assume that the bounds in P have $b_j \geq 0$ for all $j \in \{1, 2, \dots, m\}$ since if a solution satisfies an equality constraint, then it also satisfies this equality constraint by flipping the sign on both sides by multiplying by -1 .

Theorem 4.21. *Given a polyhedral set $P \subseteq \mathbb{R}^n$ defined as in (4.19) and an auxiliary polyhedral set $P_2 \subseteq \mathbb{R}_+^{2n+m}$ constructed as in (4.20). Let $\mathbf{c}, \mathbf{x} \in \mathbb{R}^n$ and $\mathbf{x}^+, \mathbf{x}^- \in \mathbb{R}_+^n$ where $\mathbf{x} = \mathbf{x}^+ - \mathbf{x}^-$, the point $\mathbf{x} \in P$ maximizes $\langle \mathbf{c}, \mathbf{x} \rangle$ if and only if $(\mathbf{x}^+, \mathbf{x}^-, \mathbf{s}) \in P_2$ maximizes $\langle \mathbf{c}, \mathbf{x}^+ - \mathbf{x}^- \rangle$, for some $\mathbf{s} \in \mathbb{R}^m$. $\square$*

Starting from an arbitrary polyhedral set P , we obtain a standard form polyhedral set P_2 that is equivalent to P . Thus, in the following, we consider only polyhedral sets in standard form.

4.5.2 Phase I: Initializing a Basic Feasible Solution